Water Storage Trends and Climate Attribution Across River Basins in Thailand

by

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ABSTRACT

Separating water-storage decline into climatic and human components requires more than a high prediction score. This study assessed terrestrial and groundwater storage anomalies (TWSA and GWSA) across Thailand's 22 principal basins using documented CSR RL06.3 mascons, a full-compartment groundwater balance, paired naturalized and historical-human WaterGAP simulations and 5,484 observed aggregates from 511 wells. The analysis combined seasonal Sen and Mann–Kendall methods with common-support product comparisons, climate-memory ablations, independent-precipitation forcing tests, additive trend uncertainty and station-panel resampling. National CSR-based TWSA declined by 7.51 mm/year through September 2024 and GWSA by 6.84 mm/year through September 2023. The contrasting COST-G/G3P chain retained a national negative direction but showed weak basin-ranking agreement: Spearman coefficients were 0.138 for TWSA and 0.027 for GWSA. Climate memory raised national held-out anomaly NSE from 0.283 to 0.890 for TWSA and from 0.039 to 0.723 for GWSA relative to current-weather Ridge models. Replacing native precipitation with CHIRPS reduced anomaly skill, exposing forcing dependence. The exact residual decomposition retained both the modeled direct-human perturbation and observation–historical-model discrepancy. The discrepancy-magnitude diagnostics were 0.938 and 0.991; these are descriptive indices, not causal human fractions. Longest-overlap residual–head correlations were 0.127 and 0.037, with broad station-bootstrap intervals. A strict balanced 2013–2022 panel gave groundwater correlation close to zero over eight usable storage years. The study demonstrates robust broad decline alongside unstable local rankings and unresolved human attribution. Its proposed contribution is an auditable credibility assessment linking processed gravity support, climate reconstruction and observation-network uncertainty, released with reproducible code and decision-support outputs.

CONTENTS

Water Storage Trends and Climate Attribution Across River Basins in Thailand **i**
CHAPTER 1 INTRODUCTION **1**
Why climate and human effects are difficult to separate 2
Objectives and proposed contribution 3
CHAPTER 2 LITERATURE REVIEW **5**
Groundwater isolation and compartment compatibility 6
Naturalized models and the meaning of attribution 7
Model skill beyond a seasonal cycle 8
Groundwater observations and network effects 9
Thai studies and the remaining research gap 10
CHAPTER 3 STUDY AREA AND DATA **12**
Input records and analysis periods 13
CSR processing and calendar audit 14
Groundwater records and representativeness 17
CHAPTER 4 METHODOLOGY **18**
Trend estimation and statistical qualification 19
Paired climate and human decomposition 20
Climate emulation and chronological evaluation 22
Held out rainfall and observed storage response 24
Groundwater panel and station bootstrap design 25
Native support and product ranking diagnostics 26
Decision support and reproducible execution 27
CHAPTER 5 RESULTS **28**
Basin trend estimates and conditional intervals 29
Product dependent magnitudes and rankings 31
Climate memory beyond seasonal prediction 35
Paired naturalized and historical human storage 38
Additive trend decomposition 39
Observation network coverage through time 42
Common period and balanced panel comparisons 43
CHAPTER 6 DISCUSSION **47**
Why strong climate reconstruction does not validate human depletion 48
Monitoring priorities and the proposed novelty 49
CHAPTER 7 CONCLUSIONS AND RECOMMENDATIONS **50**
REFERENCES 51
APPENDIX A ALL BASIN RESULTS 53
APPENDIX B REPRODUCIBILITY AND INTERPRETATION CONTRACT 54

LIST OF FIGURES

Figure 3.1 Official principal basins and analysis codes; full names appear in Appendix A. 12

Figure 3.2 National calendar-month climatologies from the retained multisource cube; the products have different physical meanings. 16

Figure 5.1 CSR-based national TWSA and full-balance GWSA; real missing satellite months remain gaps. 28

Figure 5.2 All-basin seasonal Sen slopes and conditional 95 percent bootstrap intervals; codes refer to Appendix A. 29

Figure 5.3 Basin-level CSR trend maps in millimetres per year; colours are centred on zero. 30

Figure 5.4 Common-support slopes from CSR-based and COST-G/G3P chains; the groundwater comparison changes compartments as well as gravity processing. 31

Figure 5.5 Groundwater slopes under alternative estimation structures; differing definitions prevent a single-factor interpretation. 32

Figure 5.6 Overlap of mapped native-mascon weight fractions; the matrix is not gravity-error covariance. 33

Figure 5.7 Native-weight concentration and basin area; inverse concentration is not an effective statistical sample size. 34

Figure 5.8 Controlled Ridge feature ablations evaluated in 2016–2023; boxes summarize basins and points show national context. 35

Figure 5.9 Frozen-model forcing replacement against a native-forced naturalized target; this is a sensitivity test, not rainfall-product validation. 36

Figure 5.10 Observed 2016–2023 rainfall–storage associations after development-only selection; dark bars pass basin FDR screening. 37

Figure 5.11 Observed and paired modeled storage on common support, with residual R, paired human H and discrepancy E. 38

Figure 5.12 Monthly-fixed-effect OLS slopes with shared two-year-block wild-bootstrap intervals; these point slopes are additive. 39

Figure 5.13 Conditional slope intervals under one-, two- and three-year calendar blocks; missing dates are retained. 40

Figure 5.14 Descriptive discrepancy-magnitude index D; it is not a human fraction or an independent error-budget allocation. 41

Figure 5.15 Annual coordinate-station availability by basin; values display log of one plus actual station count. 42

Figure 5.16 Residual–head relationships across quality and panel conditions; different time supports must be distinguished. 43

Figure 5.17 Eligible annual head trends and station-resampled declining fractions; head units are not storage units. 44

Figure 5.18 Primary GWS residual–head associations and coordinate-station bootstrap intervals; absent intervals indicate inadequate support. 45

Figure 5.19 Fixed-window seasonal Sen slopes; colours saturate at ±30 mm/year and full numerical values remain in the CSV. 46

LIST OF TABLES

Table 3.1 Input records and scientific roles 13

Table 3.2 Storage-compartment definitions 15

Table 4.1 Evaluation chronology 22

Table 5.1 National source and structure comparisons 31

Table 5.2 National groundwater-panel comparisons 43

Table A.1 All-basin storage, native support and head evidence 53

ABBREVIATIONS AND SYMBOLS

AIT — Asian Institute of Technology

CSR — Center for Space Research

TWSA — Terrestrial water storage anomaly

GWSA — Groundwater storage anomaly

GLDAS — Global Land Data Assimilation System

G3P — Global Gravity-based Groundwater Product

NSE — Nash–Sutcliffe efficiency

FDR — False discovery rate

OLS — Ordinary least squares

P and PET — Precipitation and potential evapotranspiration

O, C and A — Observed, naturalized and historical-human storage

R, H and E — Residual, modeled direct-human perturbation and discrepancy

D — Descriptive discrepancy-magnitude index

Head — Signed negative depth to water; not water-storage volume

CHAPTER 1
INTRODUCTION

1.1 Water storage and the research problem

Water-storage change integrates processes that are observed separately by rain gauges, land-surface sensors, reservoir records and groundwater wells. Precipitation supplies water, evapotranspiration returns part of it to the atmosphere, and routing and subsurface storage determine when the remainder becomes available. A persistent storage decline can therefore matter even where rainfall remains strongly seasonal. The practical question is whether the available observations identify a regional change, its location and the processes that explain it.

Satellite gravimetry provides a broad view of terrestrial water storage anomaly, or TWSA. It responds to changes in total water mass rather than to groundwater alone. Groundwater storage anomaly, GWSA, must be estimated by removing the other storage compartments. This subtraction introduces information from hydrological models, so the resulting groundwater series inherits their structural assumptions as well as the limitations of the gravity observations. Spatially detailed output grids do not remove that dependence.

Thailand offers an important setting for this distinction. Its principal river basins provide a useful administrative framework for comparing storage conditions, while the satellite observations integrate information across areas that can exceed individual basin boundaries. The study uses all 22 official principal basins and national context. It retains the basin definitions supplied with the project rather than redrawing boundaries to obtain stronger statistical associations or more favourable trends.

The research connects three tasks: describing storage change, evaluating a climate-related baseline and delivering results through an interactive decision-support system. These tasks require different evidence. A trend estimate describes a record; an emulator score describes a model comparison; and well agreement describes a relationship between spatially mismatched observations. Keeping these meanings distinct makes the analysis useful even where an attribution hypothesis is not confirmed.

The central proposition of this thesis is that attribution should be assessed through its sensitivity and observational support, rather than through a single high correlation. The analysis therefore evaluates product dependence, climate-memory skill, paired model discrepancy and groundwater-network uncertainty together. Its conclusions concern both the observed water-storage changes and the strength of the evidence available to interpret them.

Why climate and human effects are difficult to separate

A common attribution procedure predicts storage from meteorological variables and labels observed minus predicted storage as a human component. That interpretation is demanding. If a statistical model is trained directly on GRACE observations, its target already contains any effects of abstraction, irrigation and reservoir operation present during the training period. The fitted relationship is not automatically an estimate of a world without those activities. Adding more predictors does not resolve this identification problem.

Some predictors also respond to management. Actual evapotranspiration can increase where irrigation supplies water, and observed soil moisture can reflect both rainfall and irrigation. Such variables are useful for prediction and for describing the hydroclimatic system, but they cannot be assumed to represent an external climatic cause. This thesis consequently uses meteorological precipitation and potential evapotranspiration for the principal naturalized-model emulators and treats actual-ET and moisture products as supporting context.

A naturalized hydrological simulation supplies a more explicit counterfactual experiment. In the WaterGAP pairing used here, direct water use and artificial reservoirs are removed in one run and represented in the historical-human run. The simulations share a model and meteorological forcing. Their difference identifies the direct human perturbation represented by that model, conditional on its processes and parameters. It remains an experiment, not an independently measured pumping history.

The observation-based residual also contains whatever the historical-human run fails to reproduce. This discrepancy can include unrepresented management, climate-forcing error, groundwater-model error, compartment mismatch and gravity-processing effects. The thesis retains it explicitly. The analytical goal is to determine how much interpretive confidence remains when these alternatives are visible, rather than to force every unexplained millimetre into a human-use category.

The groundwater comparison creates a further challenge. Wells record local hydraulic head, whereas storage anomalies describe water mass over a region. Heads respond to climate, pumping and aquifer properties, and the monitored network changes over time. The study therefore asks whether apparent agreement survives detrending, climate control and changes in station composition before treating it as supporting evidence.

Objectives and proposed contribution

Objective 1 is to quantify temporal and spatial patterns of TWSA and GWSA for every principal basin in Thailand. It includes seasonal Sen slopes, original and seasonal Mann–Kendall tests, false-discovery-rate adjustment, uncertainty intervals and coverage information. The principal gravity record is the identified CSR RL06.3 product. The original project export and the documented COST-G/G3P chain provide sensitivity comparisons on common observational support.

Objective 2 is to develop and assess a climate-storage reconstruction framework, compare appropriate components with GLDAS, and examine the climate–human decomposition using groundwater observations. The study distinguishes naturalized-parent emulation, agreement with observed storage and agreement with groundwater heads. It adds controlled feature ablations, a change-of-precipitation-source experiment, held-out rainfall–storage associations and observation-network resampling to the paired hydrological experiment.

Objective 3 is to deliver an interactive decision-support system and a reproducible research package. The interface should expose basin results, missing dates, model-verification scores, observation coverage and limitations. Reproducibility includes the input subsets, executable code, dependency versions and comparison records, so that an advisor can rerun the numerical analysis without access to the original workstation.

The proposed contribution is an integrated assessment of attribution credibility across an entire national basin framework. It tests whether national decline, basin rankings, climate-memory reconstruction and groundwater agreement tell a consistent story. The mathematical decomposition itself is an accounting identity, and the statistical algorithms are established methods. The contribution lies in their controlled combination, the product and network tests, and the resulting evidence about what can be inferred in this setting.

Novelty is therefore framed as a defensible research proposition, not a claim that this is the first use of GRACE, machine learning or groundwater wells in Thailand. The literature review identifies related work and explains the differences in domain, comparison target and uncertainty treatment. Publication quality ultimately depends on the evidence, clarity and review of those differences, not on the number of methods listed.

Research questions and scope

The first research question concerns robustness: does the direction of storage change persist across documented gravity and groundwater-estimation chains, and do basin rankings remain stable? A consistent national sign would support a broad storage-change conclusion. Instability in rankings would limit local prioritization even if every individual map appeared statistically significant. These are separate questions and are evaluated on identical months and common valid cells wherever products are compared.

The second question concerns climatic information. Can precipitation, potential evapotranspiration and antecedent memory reproduce the naturalized parent model beyond a seasonal baseline? The associated expectation is that delayed hydrological response should add information about interannual anomalies. A high raw monthly score alone would not establish this, because monsoonal seasonality can produce a strong relationship with limited ability to explain unusual wet or dry conditions.

The third question concerns transfer across data sources. Does a frozen emulator remain useful when native-model precipitation is replaced with CHIRPS? This is a forcing-sensitivity experiment, not independent validation of the unobserved naturalized target. It nevertheless tests whether the reported skill depends strongly on using the same meteorological information that drove the parent simulation. Any deterioration is reported rather than repaired using the test period.

The fourth question concerns groundwater evidence. Does residual–head agreement persist when station groups are resampled and when a strictly balanced network is compared on a common period? A fixed eligibility rule alone does not guarantee a stable annual station composition. The common-period and strict-panel tests were added as an explicitly exploratory extension after this distinction became apparent in the initial network assessment.

The study is retrospective. It does not predict future climate, estimate time to aquifer exhaustion or provide measured pumping volumes. TWSA extends to September 2024, full-balance GWSA to September 2023 and paired hydrological attribution to the available 2023 model record. This scope keeps the observed and simulated periods explicit and avoids extending the counterfactual merely to fill the final observed year.

CHAPTER 2
LITERATURE REVIEW

Satellite gravimetry and spatial interpretation

GRACE and GRACE-FO infer mass redistribution from changes in the Earth's gravity field. Their hydrological value comes from observing broad storage changes that are difficult to measure through sparse compartment-specific networks. Their limitation is that the gravity inversion is regularized and spatially coarse. The estimates should therefore be interpreted through the support of the observing system rather than through the density of coordinates in a delivered raster.

CSR supplies a native geodesic mascon representation resampled to a quarter-degree grid. The provider documents corrections, a land mask and a mapping between native mascons and the convenience grid. It cautions against single-grid-point interpretation and against assuming that small basins are independently resolved. The present study follows that advice by using basin averages and by describing shared native support rather than presenting pointwise CSR significance maps (Save et al., 2016; CSR, 2026).

Product comparison remains necessary because processing choices can change amplitudes, leakage and trends. The COST-G/G3P record offers a documented contrasting chain. Both products still derive from the same satellite observing missions, so agreement is not replication by an independent physical instrument. The unidentified project export is retained as a legacy sensitivity; its close numerical relationship with CSR cannot establish its processing provenance.

Earlier work demonstrated the sensitivity of small-region estimates to processing and leakage treatment (Longuevergne et al., 2010). That literature motivates a scale-aware presentation, but it does not supply an automatic correction for the basins studied here. No gain factor is inferred by forcing agreement with wells or with a hydrological model. Such a calibration would introduce additional assumptions and could make a subsequent comparison circular.

The spatial analysis consequently has two outputs: basin-level estimates useful for screening, and explicit support diagnostics that limit their interpretation. A basin with a precise conditional slope interval can still have uncertain spatial attribution. Statistical precision within one processed record should not be confused with evidence that its boundaries isolate a physically independent gravity signal.

Groundwater isolation and compartment compatibility

Terrestrial storage combines groundwater, soil moisture, surface water, canopy water and snow or ice where present. The standard groundwater balance subtracts the non-groundwater anomalies from TWSA. In Thailand, snow is generally a minor contribution, but retaining the supplied snow and canopy terms avoids making an undocumented zero assumption. Surface water is particularly relevant because lakes, rivers and reservoirs can contribute to both seasonal variability and long-term change.

The subtraction is valid as an accounting operation only when terms share units, a reference period and spatial support. Rebaselining each component to the same valid 2004–2009 months removes offsets that would otherwise contaminate a regional balance. Applying the same valid-cell mask makes the basin means additive. The study verifies the resulting identities numerically rather than assuming that independently aggregated component means will cancel exactly.

Compatibility extends beyond regridding. A finer land-surface field can contain variability that GRACE cannot resolve. Sharifi et al. (2025) examine this problem for storage-compartment filtering and motivate matching spatial characteristics before subtraction. Their approach does not imply that every published GRACE product should receive an additional Gaussian filter. CSR explicitly advises against further empirical smoothing or scaling of its delivered mascons, which this study respects.

The primary GWSA estimate therefore retains a documented structural limitation: Noah soil storage and filtered G3P surface water have different representations. A filtered root-zone soil alternative and the native G3P groundwater product are used to assess sensitivity. The latter differs in its gravity and compartment chain, so differences cannot be assigned solely to a soil model or solely to gravity processing.

This distinction affects management interpretation. A decline that appears across alternatives is stronger evidence for direction than for magnitude. A basin whose relative position changes across alternatives should not be ranked as a uniquely severe pumping hotspot from one subtraction product. The analysis reports both the common features and the disagreements, preserving the information needed to identify where better compartment observations would be valuable.

Naturalized models and the meaning of attribution

Hydrological models represent storage through simplified processes and parameters. A naturalized simulation changes specified human interventions while retaining the model's remaining structure. WaterGAP provides a documented framework for this comparison, including historical-human and naturalized runs driven by the same forcing. The model description and dataset metadata establish the experiment; they do not guarantee that regional groundwater dynamics are reproduced accurately (Müller Schmied et al., 2024).

Let observed storage be O, naturalized storage C and historical-human storage A. The paired model difference H equals A minus C. The observational residual R equals O minus C. These quantities are equal only when A reproduces O. Defining discrepancy E as O minus A gives the exact identity R equals H plus E. The identity is useful because it prevents the part unexplained by the historical-human model from disappearing into a human-use label.

Even H has a specific interpretation. Because all series are anomalies, it describes changes in the modeled human perturbation relative to the common reference period. It does not estimate the total amount of groundwater depletion that occurred before the GRACE record or the absolute water stored in an aquifer. A small anomaly trend in H is compatible with a persistent modeled abstraction level whose influence was already present in the reference period.

Causal interpretation requires assumptions that an accounting identity cannot verify. The naturalized run must approximate the relevant counterfactual, the forcing and process representation must be adequate, and the observed and modeled storage definitions must be comparable. Errors can correlate across compartments and models. The study consequently treats the paired experiment as structured evidence and the remaining discrepancy as unresolved, rather than as an identified measurement-error term.

A useful attribution assessment should ask whether an interpretation survives a change of product, model feature set and observational network. This thesis adopts that approach. It does not assume that a larger residual proves stronger human influence, and it avoids reporting a human percentage obtained by dividing one robust slope by another. Such a percentage would mix non-additive statistics and unverified causal assumptions.

Model skill beyond a seasonal cycle

Monsoonal storage series have a strong annual cycle. A model can reproduce that cycle and achieve high correlation while missing the interannual anomalies most relevant to drought and long-term storage stress. The principal evaluation therefore reports both monthly performance and performance after removing a training-derived calendar-month climatology. The reference climatology is fitted without using the later test period.

Nash–Sutcliffe efficiency compares squared prediction error with the variance of the evaluation target. Its value depends on the benchmark and on the variability present in the chosen period. Gupta et al. (2009) and Knoben et al. (2019) show why a single efficiency score is an incomplete description of hydrological performance. This thesis pairs NSE with correlation, bias, RMSE and anomaly skill, while stating the physical meaning of each target.

Chronological separation addresses another source of optimistic assessment. Training, validation and testing occupy distinct periods, and tuning occurs within training folds. The validation period fixes ensemble membership and weights before the test is evaluated. These practices reduce temporal leakage, although they do not make an unobserved naturalized target observational truth. Structured-data validation must match the intended use of the model (Roberts et al., 2017).

Ridge regression, ElasticNet, additive splines and Random Forest provide contrasting statistical approximations to the parent model. Their role is to examine reconstructability from meteorological information. The controlled Ridge ablations hold the model family and tuning protocol constant while adding lagged and memory features. This makes it possible to identify the value of antecedent information without attributing every performance difference to algorithm complexity.

The models use weather available up to and including the reconstructed month. They are consequently hindcast or contemporaneous emulators, not operational forecasts of future storage. Independent precipitation replacement is a sensitivity experiment within that interpretation. A test score against a naturalized hydrological model, a matched-compartment GLDAS benchmark and a comparison with wells answer different questions and remain separately labelled throughout the results.

Groundwater observations and network effects

A groundwater well measures hydraulic head or depth to water at a location and within a screened interval. It can provide independent information about aquifer response, but it does not directly observe the integrated water mass represented by GWSA. Converting head change to storage requires defensible aquifer storage coefficients and information about confinement. These are unavailable for the present Thai well records, so the study compares signed head patterns and trends without inventing a conversion factor.

A regional head index also depends on the monitoring network. Several wells may occupy the same coordinate group, and their records may begin in different years. Simply averaging all available wells can change the represented population over time. The study first combines colocated wells and gives equal weight to coordinate station groups. It then reports how many groups actually contribute to each annual index.

Annual observations cannot be treated as monthly measurements merely because the satellite record is monthly. Most retained Thai observations are provider aggregates for annual intervals, often based on few measurements. The comparison therefore uses annual storage means with a minimum observed-month requirement and excludes incomplete endpoint years. Missing GRACE months and mission-gap years are not filled to obtain a more convenient sample.

Correlation is informative only with its conditioning choices visible. Shared seasonality, climate variability and broad trends can create positive relationships between heads and a storage residual. Raw correlations are supplemented by climate-controlled, detrended and contiguous-year-change comparisons. Partial correlation is still descriptive, because the naturalized model is not a direct observation of the true climatic head component.

Network uncertainty is examined by resampling coordinate groups and by comparing observation panels. A fixed eligibility rule permits some annual missingness; a strict balanced panel requires all ten annual observations in 2013–2022. The all-well common-period comparison separates a change in time window from a change in monitored population. These tests make the groundwater evidence more transparent, including cases where an apparently stronger correlation does not survive a fair comparison.

Thai studies and the remaining research gap

Previous research has already combined GRACE, land-surface modeling and groundwater observations in the Chao Phraya system. Abhishek et al. (2021) examined storage dynamics and hydroclimatic extremes, while Abhishek et al. (2022) investigated a multisource water budget in the upper basin. These studies establish relevant regional context. The present work does not claim that storage separation or hydroclimatic analysis is new to Thailand.

Sharma et al. (2025) developed a GRACE-based aquifer reliability and resilience framework and reported strong in situ agreement in their Chao Phraya sub-basin application. Their result does not imply that the different well population, basin system and comparison periods used here must show the same correlations. Validation statistics should be interpreted with the exact head records, storage definitions and sampling support that produced them.

Basin terminology requires particular care. The official 22-basin framework separates Ping, Wang, Yom, Nan and other principal basins from the basin labelled Chao Phraya. A study of the greater Chao Phraya drainage system can therefore cover a larger area than the official Chao Phraya polygon used in this analysis. Directly comparing slope magnitudes without checking those boundaries would confound domain definition with hydrological change.

The remaining gap addressed here is the joint assessment of several inferential links across every principal basin. The study tests gravity-product dependence and shared native support, checks climate memory beyond seasonality, retains observation–model discrepancy and examines the observation network used for groundwater agreement. Its public interface and portable code expose those tests alongside the basin estimates.

The resulting novelty claim is intentionally specific: the research offers an auditable national attribution-credibility assessment with controlled product, forcing and network sensitivities. A claim of priority would require a broader systematic literature search than the focused review used here. The defensible contribution is the completed experiment and its findings, including negative validation evidence that would be hidden by reporting only the most favourable model or panel.

Synthesis of the analytical framework

The literature points to three distinct levels of evidence. The first is measurement and accounting: identify the gravity product, preserve its time support, harmonize units and references, and verify the compartment balance. The second is model evaluation: determine whether meteorological information reconstructs a documented naturalized simulation and whether that simulation agrees with observed storage. The third is independent consistency: examine groundwater observations with their spatial and temporal support retained.

None of these levels replaces the others. A strong model-emulation result does not repair an inaccurate parent model. A significant basin trend does not establish an independent spatial signal. A positive head correlation does not convert a total-storage residual into a human pumping volume. The research design makes these distinctions explicit in both the computational workflow and the final decision-support interface.

The controlled tests are arranged to identify where interpretations are sensitive. The product audit holds observational support constant while changing the processed storage chain. The feature ablation holds the statistical family and data split constant while changing antecedent information. The forcing stress test freezes the fitted model while changing precipitation source. The common-period head comparison holds time support constant while changing the station panel.

The paired hydrological experiment supplies the central decomposition. It is evaluated with a common-date, monthly-fixed-effect linear trend partition and shared-multiplier uncertainty analysis. That linear partition is kept separate from the robust Sen summaries. This separation preserves exact additivity where it is mathematically justified and avoids creating spurious fractions from statistics that do not add.

The practical output is a set of evidence-qualified basin assessments. A user can identify a declining record, inspect the dependence of its magnitude and ranking on the product, examine climate reconstruction and see whether groundwater observations provide consistent support. The framework can guide further monitoring and model improvement without claiming that the currently available data resolve every component of human influence.

CHAPTER 3
STUDY AREA AND DATA

Study area and official basin framework

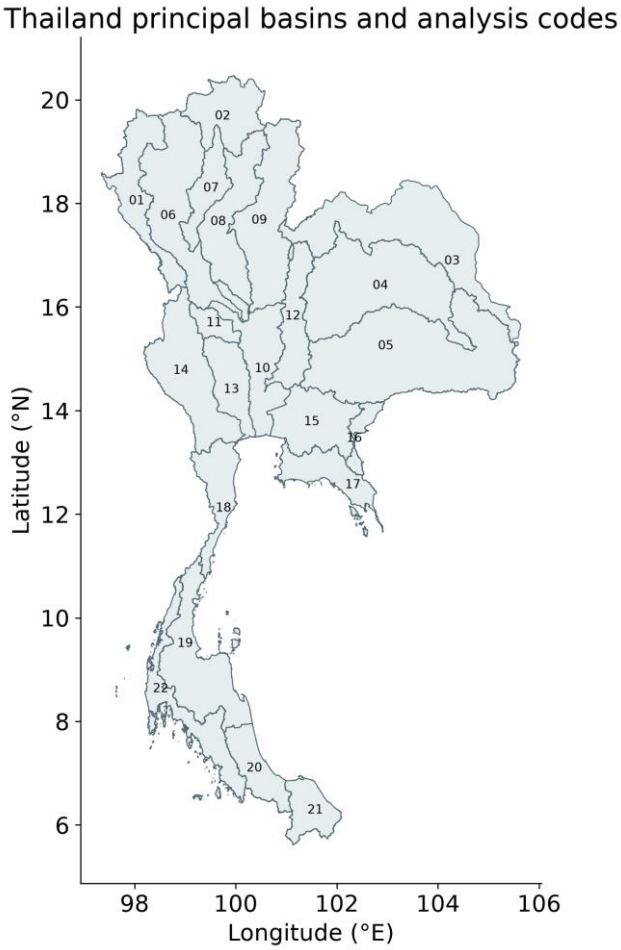


Figure 3.1 Official principal basins and analysis codes; full names appear in Appendix A.

The spatial framework contains 22 principal basins and separately coded island features assigned to their parent basin for the revised analysis. The retained polygon–grid intersection area totals approximately 515,230 km² for national context. This is the analytical intersection-area total, not a replacement estimate of Thailand's official national area.

The official Chao Phraya principal basin is not synonymous with the entire greater Chao Phraya drainage system used in some earlier studies. The distinction is retained when interpreting local literature or comparing basin slopes. No basin boundaries are selected to maximize a trend or correlation. National context provides a larger-scale comparison, while all individual principal basins retain the CSR provider's small-basin caution.

Input records and analysis periods

Table 3.1 Input records and scientific roles

Record	Period used	Role
CSR RL06.3	Apr 2002–Sep 2024	Primary gravity storage
GLDAS Noah	Apr 2002–Sep 2024	Soil, canopy, snow
G3P v1.12	Apr 2002–Sep 2023	Surface water and alternatives
WaterGAP 2.2e	1981–2023	Naturalized/historical pairing
Multisource cube	2003–2024	CHIRPS and ancillary context
Thai GROW observations	2004–2024 annual; 2018–2023 monthly	Groundwater consistency

The study combines an identified satellite-gravity baseline, land-surface storage components, naturalized and historical-human hydrological simulations, multisource hydroclimate fields and actual groundwater observations. The roles are deliberately different. CSR supplies the principal observed mass-anomaly record. GLDAS Noah and G3P supply non-groundwater compartments or structural sensitivities. WaterGAP supplies the paired counterfactual experiment and the target for climate emulation.

The main calendar spans April 2002 to September 2024, but availability differs by product. The documented CSR subset contains 237 observed monthly solutions over that interval. Surface-water availability limits primary GWSA to September 2023, producing 225 observed GWS months. No interpolation is introduced between the GRACE and GRACE-FO missions. The common-support source audit uses fewer months where the legacy and G3P records require it.

Each supplied or downloaded input is accompanied by a source inventory or provenance record. The extended package adds CSR release information, download checksums, nominal-month assignments and native mapping. This improves the baseline provenance while preserving a clear distinction between the new primary analysis and the earlier unidentified export.

CSR processing and calendar audit

The CSR RL06.3 all-corrections NetCDF is downloaded from the provider together with its land mask and native-mascon mapping. A compact Thai-buffer subset is retained, and the original source checksum is recorded. The primary analysis uses the delivered corrected mass field in centimetres, converted to millimetres. It does not undo provider corrections or add an empirical gain factor.

The source uses an uppercase Units attribute for time and thickness. The acquisition code explicitly interprets the declared time epoch rather than assuming that a generic CF decoder has converted the floating-point time coordinate. This check is important because a time-selection operation can otherwise return a numerically plausible but incorrect subset. The complete monthly assignment audit is saved with the input data.

Two source midpoints fall in the preceding calendar month of their nominal GRACE solution: the second October 2011 entry is assigned to November 2011, and the late-April 2015 entry to May 2015. They are retained as distinct observations. The audit records the original midpoint and the assigned month, while the source averaging bounds remain available. These assignments follow the established special-month issue described in GRACE time-processing guidance.

The CSR provider land mask is applied before basin aggregation. This reduces the use of ocean-coded cells at coastlines but does not constitute a full leakage correction. Missing observations are reindexed onto a complete calendar as missing values. The resulting time series therefore distinguishes absence of a satellite solution from a measured anomaly close to zero.

The data audit is part of the scientific method rather than an incidental software step. Misread units, merged special months or silently filled mission gaps can influence an apparent trend, lag or validation sample. Explicit checks and retained metadata make those choices inspectable and allow a reviewer to separate data-handling decisions from later hydrological interpretation.

Storage compartments and structural alternatives

Table 3.2 Storage-compartment definitions

Term	Primary representation
TWSA	CSR corrected mass anomaly
SMSA	Four GLDAS Noah soil layers
SWSA	G3P modeled surface-water anomaly
CWSA and SWEA	GLDAS canopy and snow
Soil sensitivity	Filtered root-zone alternative

Primary soil storage is the sum of the four GLDAS Noah soil-moisture layers. Canopy interception and snow-water-equivalent storage are retained as separate terms, and their sum with soil storage is checked against the supplied Noah land-storage total. The source units are interpreted as equivalent water depth. This component check guards against a scale or band-selection error in the regional conversion.

Surface-water anomalies are obtained from the documented G3P surface-water chain, which represents rivers, lakes and reservoirs through its underlying modeled fields. This is not a replacement for independently observed Thai reservoir histories. The distinction matters in attribution, because the surface-water model may not reproduce actual operations or the full volume change associated with specific infrastructure.

The filtered root-zone moisture alternative changes the soil contribution to the groundwater subtraction while retaining the CSR gravity baseline and the remaining compartments. It tests sensitivity to soil representation and filtering. It should not be interpreted as a correction that proves either soil product is right. The products represent different depths and spatial characteristics, so their differences are a structural uncertainty.

Hydroclimatic context and seasonal structure

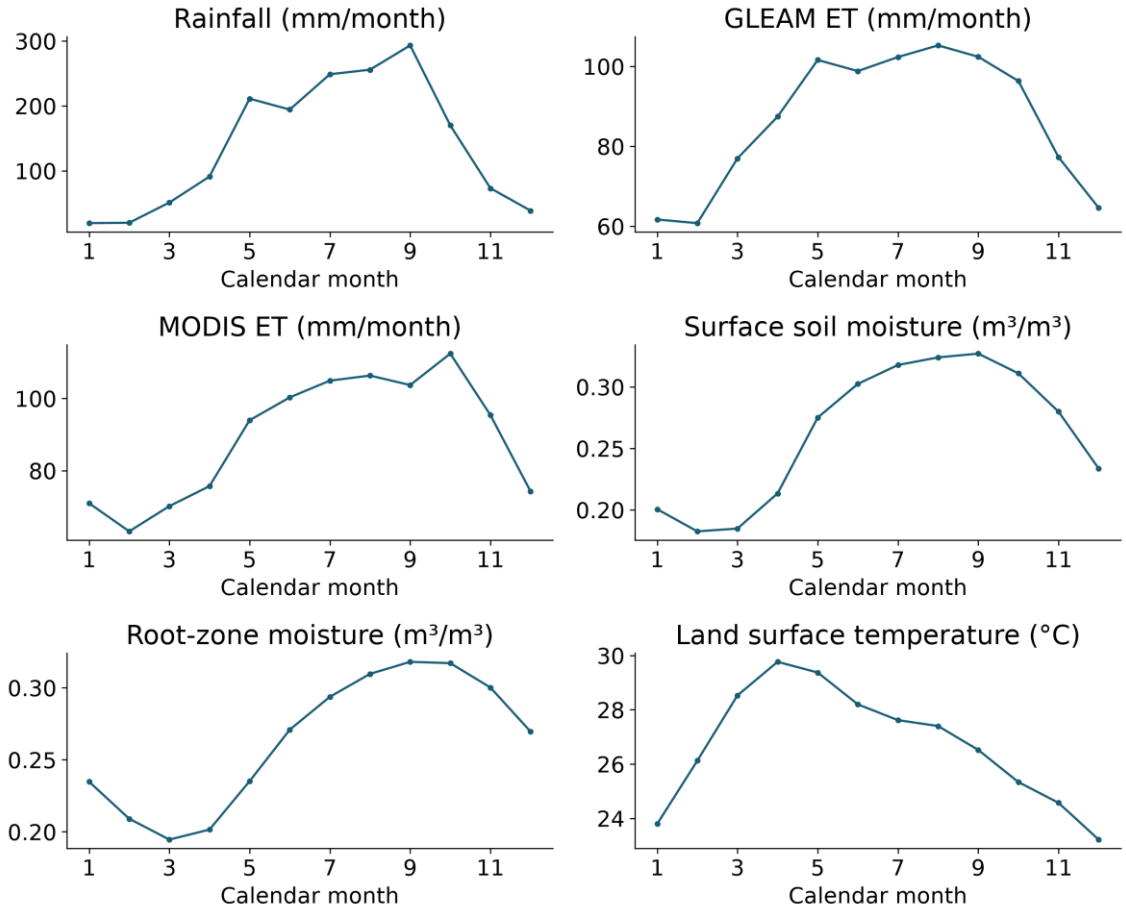


Figure 3.2 National calendar-month climatologies from the retained multisource cube; the products have different physical meanings.

Rainfall, actual evapotranspiration, soil moisture and land-surface temperature describe complementary parts of the regional seasonal cycle. The plotted climatologies summarize available 2003–2024 observations in the harmonized cube; they are not used to fill missing gravity or groundwater measurements.

The two ET records and the soil products are retained as context rather than assumed to be external climatic causes. Their response can include management effects, and their structural definitions differ. CHIRPS supplies the independent precipitation forcing and the observed rainfall–storage test. The main naturalized-model emulators use meteorological P and potential ET, with training-derived seasonal references.

Groundwater records and representativeness

The retained groundwater tables contain actual observed aggregates from the GROW dataset, attributed to the Thai source through the package metadata. Six Thai source rows without observed depth were removed during normalization. Filled groundwater columns are not used. The extraction preserves identifiers, coordinates, aggregation interval and provider quality flags, allowing the well population to be audited independently of the storage analysis.

Depth below ground is positive downward. The analysis therefore defines signed head as negative depth: rising signed head indicates a shallower water table or higher hydraulic head under the supplied convention. Unitless per-well indices are constructed from the observed head series for descriptive regional comparison. This normalization is not used as a climate-model predictor or as a basis for converting head to groundwater mass.

Co-located wells are combined into coordinate station groups before regional averaging. This avoids allowing several wells at one location to dominate a basin index. Annual availability is still uneven, and network density changes materially through the record. The coverage table reports observed station counts by basin, year and panel, so a smooth regional line cannot hide a change in the population it represents.

Aquifer names, screened intervals and storage coefficients are not sufficiently documented for aquifer-specific volumetric validation. Some package attributes describe mapped environmental conditions; these are not a substitute for verified well construction and hydrostratigraphic information. The analysis consequently reports pattern consistency, well-head trend direction and observation support rather than a calibrated groundwater storage volume.

The strict-panel experiment requires ten of ten annual observations during 2013–2022 and compares it with the all-well index over the same calendar window. Only years with adequate observed storage contribute to the correlation. Mission-gap years remain unavailable even where wells were observed. This deliberately reduces sample size in exchange for a more interpretable comparison of station composition.

CHAPTER 4
METHODOLOGY

Spatial aggregation and the groundwater balance

$$GWSA = TWSA - SMSA - SWSA - CWSA - SWEA$$

Basin aggregation uses the supplied polygon–grid intersection areas rather than a simple selection of grid-cell centres. For each month, the weighted mean is the sum of valid storage values multiplied by their intersection areas, divided by the sum of valid intersection areas. A coverage fraction records the valid area relative to the complete basin weight. Months below the prespecified 50 percent threshold are left missing.

This procedure retains coastal and boundary intersections and makes the role of missing cells explicit. It does not make the underlying gravity information spatially finer. The same intersection grid is used for CSR, the compartment fields and the paired model comparison. Coarse model cells are replicated to aligned convenience-grid locations using nearest-parent assignment, which adds no independent spatial information.

The groundwater balance is calculated at the common grid and reference support before its basin series is exported. All terms are in millimetres of equivalent water height. A common 2004–2009 valid-month reference is used for the component means. This gives an exact accounting relationship for the retained data and separates reference effects from physical storage differences.

The analysis checks the grid-level groundwater identity and the basin-level identity after weighting. Shared support is essential for both. A sum of independently calculated basin averages can fail to equal the total when one compartment has a different coastal mask or temporal coverage. The exported TWSA_GWS_reference field is therefore the appropriate total to use when reproducing the groundwater equation.

Coverage is a screening condition, not an uncertainty estimate. A month can exceed the threshold while remaining affected by leakage, model error or incomplete basin resolution. Those limitations are examined through the product audit and native-support diagnostics. The analysis does not replace them with an arbitrary confidence weight or infer a gain factor from the desired magnitude of groundwater agreement.

Trend estimation and statistical qualification

Seasonal Sen slopes are calculated from pairs of observations in the same calendar month and divided by their actual separation in years. The median of those pairwise slopes provides a robust summary that is less sensitive to individual extremes than ordinary least squares. Missing calendar months remain missing, and the calculation does not compress a mission gap into a shorter elapsed interval (Sen, 1968).

Original and seasonal Mann–Kendall tests are retained because they answer related but different questions. The seasonal test accommodates the recurring annual cycle through within-season comparisons, while the original test provides a familiar sensitivity. The study also reports annual comparisons where adequate observed months exist. The tests are interpreted with the serial-dependence limitations identified by Hirsch et al. (1982) and Hirsch and Slack (1984).

False-discovery-rate adjustment is applied within each storage variable across the 22 basin tests. It reduces the risk of treating a large set of unadjusted p-values as independent discoveries, but it does not repair weak spatial resolution or prove independence among neighbouring basins. Very small numerical p-values are reported as thresholds rather than as a literal probability of zero (Benjamini and Hochberg, 1995).

Primary Sen-slope intervals use 300 calendar-year residual-block resamples. These quantify sensitivity within the processed record. They do not propagate the full gravity inversion covariance, errors in every compartment or alternative hydrological-model structures. The separate source and soil comparisons therefore remain necessary even when a conditional interval excludes zero.

Trend-window sensitivity uses fixed windows chosen to distinguish the full common record, early and later periods, and the intervals before and after the mission gap. Short-window slopes are descriptive and are not interpreted as acceleration estimates. A few wet or dry years can change a five-year slope substantially. The study reports that variability so that a long-record decline is not mistaken for a monotonic year-by-year trajectory.

Paired climate and human decomposition

$$R = O - C = (A - C) + (O - A) = H + E$$

The attribution baseline is the published naturalized WaterGAP simulation rather than a prediction trained directly on GRACE. Naturalized and historical-human storage are placed on the analysis grid and rebaselined over the same observed 2004–2009 reference cells and months. Attribution is calculated only where the observed and both modeled fields are available, and every component uses that same footprint.

The paired modeled human perturbation is H equals A minus C, where A is historical-human storage and C is naturalized storage. The observation-based residual is R equals O minus C. Their difference E equals O minus A is retained as discrepancy. The identity R equals H plus E is checked both in the monthly tables and in the additive trend calculation. Neither term is silently reallocated to another category.

This design has a clear advantage over naming a fitted empirical relationship natural. The interventions removed from the naturalized model are documented, and the model pairing controls the forcing and much of the numerical structure. It also has a clear limit: agreement between the two simulations is not independent evidence that the actual climate-only trajectory is known. Groundwater processes and management can still be imperfectly represented.

The model-only record is retained for climate emulation, including the long warm-up and training periods before GRACE. The observed attribution record is shorter and contains real observational gaps. These records are not interchangeable. A test score for reconstructing continuous naturalized storage is not calculated on the same target as a comparison between modeled groundwater and the derived GRACE groundwater balance.

All quantities are anomalies relative to the reference. The paired difference therefore describes changes in modeled human influence relative to that period. The study does not infer cumulative depletion since the onset of development. It also does not extrapolate the naturalized model into 2024 merely because observed TWSA continues into that year. These choices keep both the physical meaning and the temporal support of the decomposition explicit.

Additive trends and conditional uncertainty

$$D = |\beta_E| / (|\beta_H| + |\beta_E|)$$

A robust Sen slope is a median operator. The Sen slope of a sum therefore need not equal the sum of the component Sen slopes. Dividing a residual Sen slope by an observed Sen slope would not provide a mathematically exact trend partition, even before causal assumptions are considered. The study keeps robust component summaries separate from an explicitly additive linear calculation.

The additive calculation fits ordinary least squares with a time trend and calendar-month fixed effects. All six component series use exactly the same observed dates and the same design matrix. Linearity then preserves the identities between the fitted point slopes. The output includes observed, naturalized, historical-human, paired human, residual and discrepancy coefficients in millimetres per year.

Conditional intervals use a calendar-block wild regression bootstrap. The fitted residuals are multiplied by shared random signs for all components within a calendar block. Using the same multiplier across the component vector preserves the accounting relationships in every replicate. Original observation dates and missingness are retained. The analysis uses 999 replicates and repeats the calculation with one-, two- and three-year blocks.

This is an unstudentized, percentile multiplier assessment conditional on the observed record and chosen regression. It is motivated by clustered-error inference but is not presented as a complete inversion-error model or a universal small-sample confidence procedure (Cameron et al., 2008). The limited number of calendar blocks and the sensitivity to block length remain part of the interpretation.

A discrepancy-magnitude diagnostic compares the absolute discrepancy slope with the sum of the absolute modeled-human and discrepancy slopes. Its purpose is to show whether the residual interpretation is dominated by the unexplained part of the paired experiment. It is bounded between zero and one, but it is not a causal share, a fraction of observed depletion or a partition of independent error variances. The report labels it accordingly.

Climate emulation and chronological evaluation

Table 4.1 Evaluation chronology

Role	Years	Target use
Warm-up	1981–1990	Antecedent climate memory
Train	1991–2009	Fit and internal tuning
Validation	2010–2015	Freeze model selection
Test	2016–2023	Held-out evaluation

The naturalized parent model supplies a consistent training target for climate emulation. Meteorological precipitation and potential evapotranspiration provide the main inputs. Causal lagged values represent antecedent forcing, exponentially weighted memories summarize persistence, and annual and semiannual harmonics represent recurring seasonality. No observed GRACE lag, well head or unrestricted calendar-year trend is included in the principal feature set.

The chronology is fixed before evaluation: 1981–1990 supplies warm-up, 1991–2009 training, 2010–2015 validation and 2016–2023 testing. Within-training chronological folds select regularization or tree settings. Scaling and feature transformations are fitted within each training fold. This prevents the later target distribution from determining training transformations or tuning choices.

Ridge, ElasticNet, additive splines and Random Forest provide the original ensemble candidates. Validation identifies up to three members that improve on the training climatology and retain positive anomaly skill. Inverse validation-error weights are frozen before testing. The ensemble is evaluated once on the held-out parent-model target, with its target and period stated beside every score.

The retained base analysis also compares naturalized WaterGAP soil, canopy and snow storage with the same GLDAS Noah compartments. National agreement over 261 months is r=0.969, NSE=0.738 and RMSE=52.49 mm. This matched-compartment benchmark is distinct from full TWS and GWS validation: groundwater and surface water are excluded, climatic drivers may be shared, and GLDAS already contributes to the GWS subtraction.

Forcing sensitivity and memory comparisons

An emulator trained and evaluated with the forcing used by its parent simulation benefits from a consistent input–target relationship. That relationship is appropriate for emulation, but its strength can be mistaken for independent physical validation. The forcing experiment therefore freezes the full Ridge model and replaces its precipitation input with CHIRPS while leaving potential evapotranspiration and the naturalized target unchanged.

CHIRPS monthly precipitation is converted to mean millimetres per day using the actual number of days in each calendar month. WaterGAP meteorological rates are converted directly to mean millimetres per day, avoiding an assumption that the source 360-day calendar represents physical monthly totals. The products are thus compared through a common rate convention rather than through incompatible accumulation lengths.

The experiment includes unadjusted CHIRPS and a training-only calendar-month mean adjustment. The adjustment is estimated from the available 2003–2009 overlap and is not recalculated in the test period. It addresses a limited mean-seasonal difference, not every distributional or process difference. A change in score after adjustment is reported whether it improves or worsens performance.

Memory gain is assessed using the difference in test squared errors between the current-weather and memory feature groups. The two predictions are compared on the same months. A circular two-year paired bootstrap resamples the eight test years to describe the stability of that error difference. With only eight years, the interval is a conditional comparison rather than a broad estimate of performance under all future climates.

The source-change result is interpreted as covariate-shift sensitivity against a native-forced parent target. CHIRPS is not expected to reproduce that target perfectly, and a reduced score is not proof that CHIRPS is less accurate as rainfall. The useful question is how much the emulator's apparent skill depends on forcing consistency. This separates an approximation success from an independent observational claim.

Held out rainfall and observed storage response

The observed rainfall–storage analysis provides a complementary test using CHIRPS and CSR-derived storage. It asks whether precipitation accumulation and delay selected in an earlier period remain associated with later storage anomalies. This is distinct from emulating WaterGAP and does not require that the naturalized parent model reproduce observed groundwater accurately.

Candidate precipitation accumulation windows are one, three, six and twelve months, with delays from zero to six months. The best positive development-period relationship is selected using 2003–2015 only. Seasonal means and a linear trend in storage are estimated in that development period and applied consistently to the later data. The selected accumulation and delay are then frozen for the 2016–2023 comparison.

A held-out Pearson correlation describes the later relationship. Its interpretation remains limited: an accumulation window and delay summarize an empirical response, not an aquifer residence time, recharge travel time or unique climatic cause. Groundwater pumping can respond to rainfall deficits, so climate and management pathways can remain correlated even in a deseasonalized, detrended analysis.

The null comparison permutes whole precipitation years in the test period while retaining within-year order and the actual missing storage positions. This preserves more temporal structure than shuffling individual months, but it does not preserve every interannual dependence. The analysis uses 999 permutations and adjusts basin test p-values within each storage type. National context is reported separately from the 22-basin family.

The test is deliberately modest in its claim. A transferable rainfall–storage association supports a role for hydroclimatic variability in observed anomalies. It does not establish that a simulated naturalized trend is correct, nor that the remaining trend is entirely anthropogenic. The result is interpreted alongside the paired-model discrepancy, product sensitivity and groundwater checks rather than being used as a shortcut around them.

The reservoir extension uses MSEA-Res screened scene estimates and, separately, provider-interpolated daily storage. A fixed pre-2004 panel is summed only when every reservoir is represented, converted from km^3 to basin-equivalent mm and referenced to available 2004–2009 months. Matched-month correlations remove calendar-month effects and linear trend; 499 whole-year cluster resamples describe conditional uncertainty. Reservoirs provide surface-water consistency evidence, not observed human-only storage.

Groundwater panel and station bootstrap design

The primary groundwater index is the mean of coordinate-station median standardized heads for each observed year. Standardization is descriptive over each well's observed record and is not used to train a predictive climate model. Regional comparisons preserve the number of station groups contributing to each year. Co-location grouping prevents multiple wells at one coordinate from being treated as independent spatial observations.

Five observation conditions are evaluated. The longest-overlap index retains provider flags. A quality sensitivity excludes flagged observations. A fixed-eligibility panel retains wells with at least eight observations in 2013–2022 but keeps their wider histories. An all-well common-period index restricts the time window to those ten years. A strict balanced panel requires observations in every one of those years and uses that same window.

The last two conditions are essential for interpretation. Comparing a longest-overlap series with a restricted panel can change both station composition and effective time support. Comparing the all-well and strict panels on the same storage years isolates the panel difference more clearly. It still does not create an aquifer-representative random sample, because the well locations and original monitoring purposes are not controlled by this study.

Station-bootstrap intervals resample coordinate groups with replacement and retain each selected group's whole time series. They use 499 replicates and quantify sensitivity to the monitored spatial population conditional on the available years. They are not temporal confidence intervals for correlation and do not propagate storage-product uncertainty. A narrow station interval can therefore coexist with a small temporal sample and weak causal evidence.

Raw, climate-controlled and detrended associations are accompanied by correlations of changes between consecutive observed years. Nonconsecutive years are never treated as one-year differences. Well-level head trends require at least eight annual observations and an eight-year span; station-level direction summaries then prevent colocated wells from dominating the national count. No head trend is converted into a water-storage or pumping rate.

Native support and product ranking diagnostics

The CSR mapping file permits the convenience-grid intersection weights to be grouped by native mascon identifier. Each basin is described by the number of native mascons it touches, the largest native weight and an inverse concentration measure. The latter equals one divided by the sum of squared native weight fractions. It describes how concentrated a basin's representation is among native elements.

This concentration count is not an estimate of independent observations. Regularization couples gravity information, and the true averaging kernels or full error covariance are not available in this analysis. Treating the count as a statistical sample size would be unjustified. Its value is diagnostic: a basin represented mainly by one or two native weights should not be interpreted as finely resolved merely because many quarter-degree cells intersect it.

A shared-support matrix sums the minimum of the native weight fractions for each pair of basins. A value of one on the diagonal is an accounting property; positive off-diagonal values identify overlap in the mapped representation. The matrix is not a leakage correction, spatial correlation matrix or gravity-error covariance. It makes one source of spatial dependence visible without pretending to solve it.

Product rankings are compared with Spearman correlation on common-valid source tables. The corresponding p-values are retained only as descriptive calculations because basin dependence weakens an independent-spatial-replication interpretation. Direction agreement, median slope differences and the ranking coefficient carry the substantive comparison. The analysis does not choose the product that produces the strongest ranking agreement or the largest decline.

The provider's small-basin caution is retained as a separate flag. All 22 principal basins fall below the cited approximate area threshold, while national context has a larger support. This does not imply that every basin result is useless or that the threshold is a binary resolution theorem. It means the basin estimates are screening information requiring regional corroboration and cannot support precise local pumping rankings on their own.

Decision support and reproducible execution

The decision-support system connects basin selection with storage time series, trend summaries, climate-model information and groundwater evidence. Its scientific function is transparency: users should be able to see the record, the counterfactual comparison and the limits of the available observations together. A map alone cannot communicate whether a basin rank is product-dependent or whether an attractive model score refers to a parent model rather than independent storage.

The original application and the extended research outputs are versioned separately. The extended study uses CSR as its identified primary baseline and exposes the source audit, memory tests, discrepancy diagnostic and network-panel comparisons. Earlier outputs remain accessible as historical work. This avoids silently changing the interpretation of an earlier downloaded table while giving readers a clear current analysis path.

The research package includes compact Thai input subsets, dependency versions, scripts, provenance and frozen reference tables. The principal analysis can be rerun from a fresh checkout without the author's drive structure. New outputs are written to a separate working directory, and automated checks compare the regenerated scientific tables with reference values using explicit tolerances.

Reproduction checks address executable consistency, not the truth of a hydrological hypothesis. A successful run confirms that the stated inputs and methods produce the reported numbers. It cannot establish missing aquifer properties, eliminate gravity leakage or prove that the naturalized simulation is the true unobserved world. These limits remain visible in the code documentation, report and interface.

The interface is intended for comparison, observation-gap identification and research review. It should not automatically prescribe pumping reductions from an unexplained residual. A useful next action can instead be to improve monitoring in a poorly supported basin, recover reservoir operation records or test another naturalized model. Linking each result to its evidence makes that distinction operational rather than leaving it only in the discussion text.

CHAPTER 5
RESULTS

National storage change

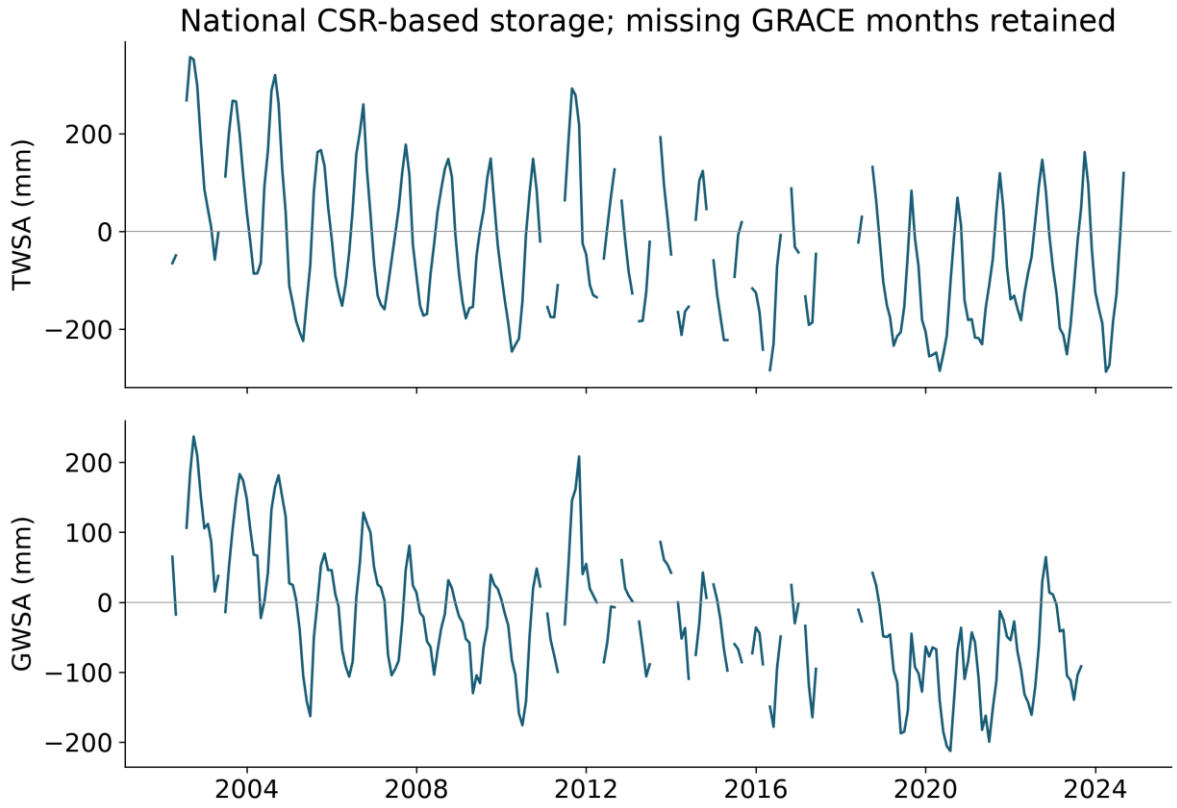


Figure 5.1 CSR-based national TWSA and full-balance GWSA; real missing satellite months remain gaps.

The primary CSR record gives a TWSA seasonal Sen slope of −7.51 mm/year, with a 300-resample conditional interval of −9.79 to −5.68. Derived GWSA declines at −6.84 mm/year, with an interval of −8.77 to −5.11. TWSA contains 237 observed monthly solutions through September 2024; GWSA contains 225 through September 2023.

These are robust summaries of processed storage records, not constant annual losses in every year. The seasonal cycle and shorter-period recoveries remain visible. GWSA is a compartment-derived estimate with model and filtering uncertainty, and its shorter endpoint follows surface-water availability. The intervals do not contain every source of product or structural error.

Basin trend estimates and conditional intervals

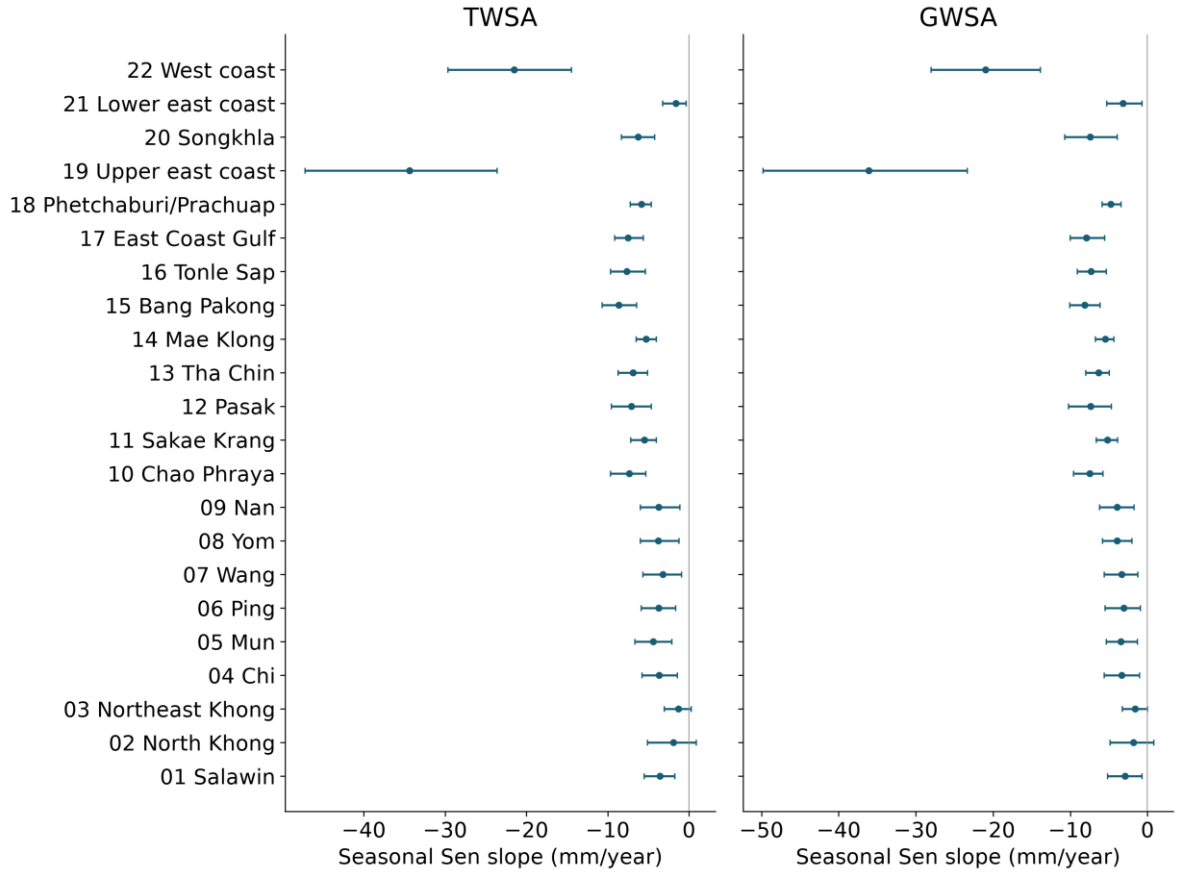


Figure 5.2 All-basin seasonal Sen slopes and conditional 95 percent bootstrap intervals; codes refer to Appendix A.

All principal basins have negative primary full-record slope estimates for both storage types. Their magnitudes vary substantially, with several southern coastal basins showing particularly large declines. The interval calculations describe uncertainty within the selected processed record and should not be read as a product-independent ordering of basin stress.

The common-source audit, soil alternative and native-support analysis are therefore needed alongside this forest plot. A basin interval can exclude zero while its relative position remains sensitive to another processing chain. The national result is supported over a larger area; local interpretations require additional compartment and aquifer evidence.

Spatial distribution of basin storage trends

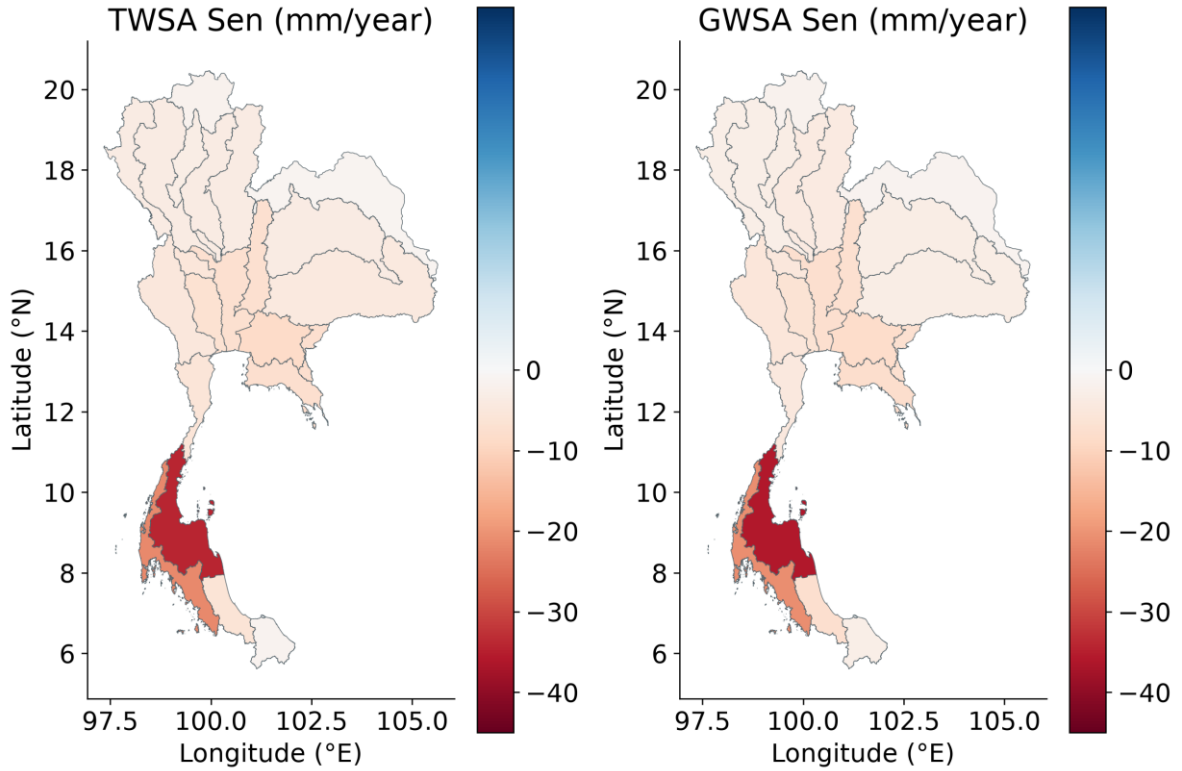


Figure 5.3 Basin-level CSR trend maps in millimetres per year; colours are centred on zero.

The maps display regional screening estimates rather than independent quarter-degree groundwater observations. Basin polygons are the reporting framework, while the gravity information is spatially broader and regularized. The large southern magnitudes deserve attention together with the coastline and support diagnostics.

A full-compartment subtraction has been used, so the groundwater pattern is not obtained by ignoring surface water or canopy storage. Nevertheless, represented soil depth, surface-water filtering and coastal support remain structural uncertainties. The maps should guide investigation and comparison with independent records rather than directly prescribe a basin-specific pumping reduction.

Product dependent magnitudes and rankings

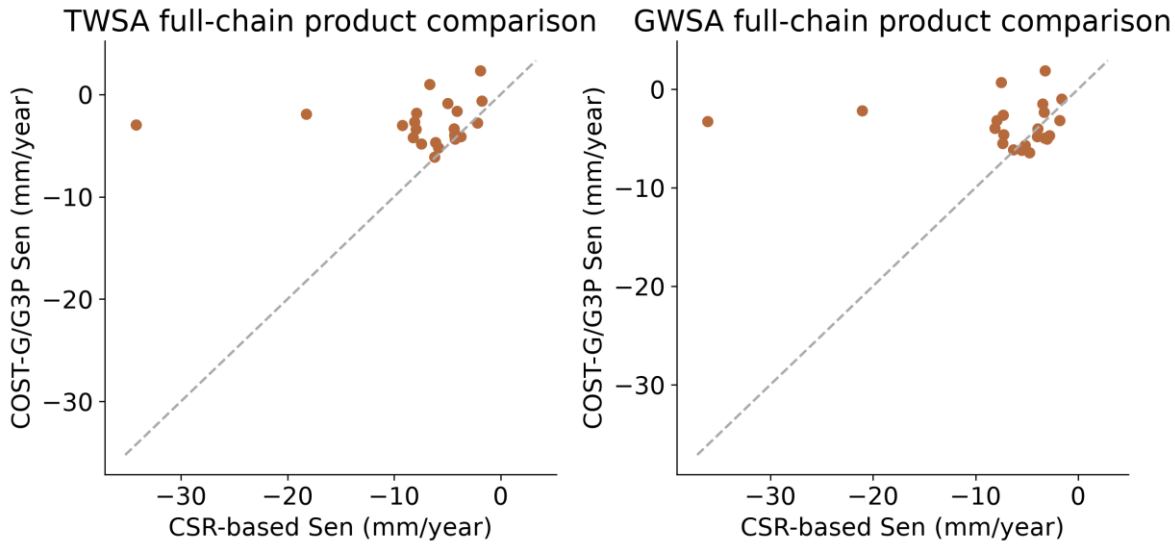


Figure 5.4 Common-support slopes from CSR-based and COST-G/G3P chains; the groundwater comparison changes compartments as well as gravity processing.

National decline persists in both documented chains, but magnitude and local ordering differ. On the 223-month common source support, TWSA Sen slopes are approximately −7.67 for CSR and −2.56 mm/year for COST-G/G3P. Groundwater slopes are −6.88 and −3.45 mm/year, respectively. These contrasts are not removed by a common anomaly reference.

Across the 22 basins, rank agreement is weak: Spearman coefficients are 0.138 for TWSA and 0.027 for GWSA. The legacy export agrees closely with CSR but has unresolved provenance and is not counted as independent replication. Agreement in national direction is consequently stronger than agreement in a basin priority list.

Table 5.1 National common-support Sen slopes in millimetres per year

Processing chain	TWSA	GWSA
CSR-based	-7.67	-6.88
COST-G/G3P	-2.56	-3.45
Legacy unidentified	-7.68	-6.86

Groundwater product and soil representation

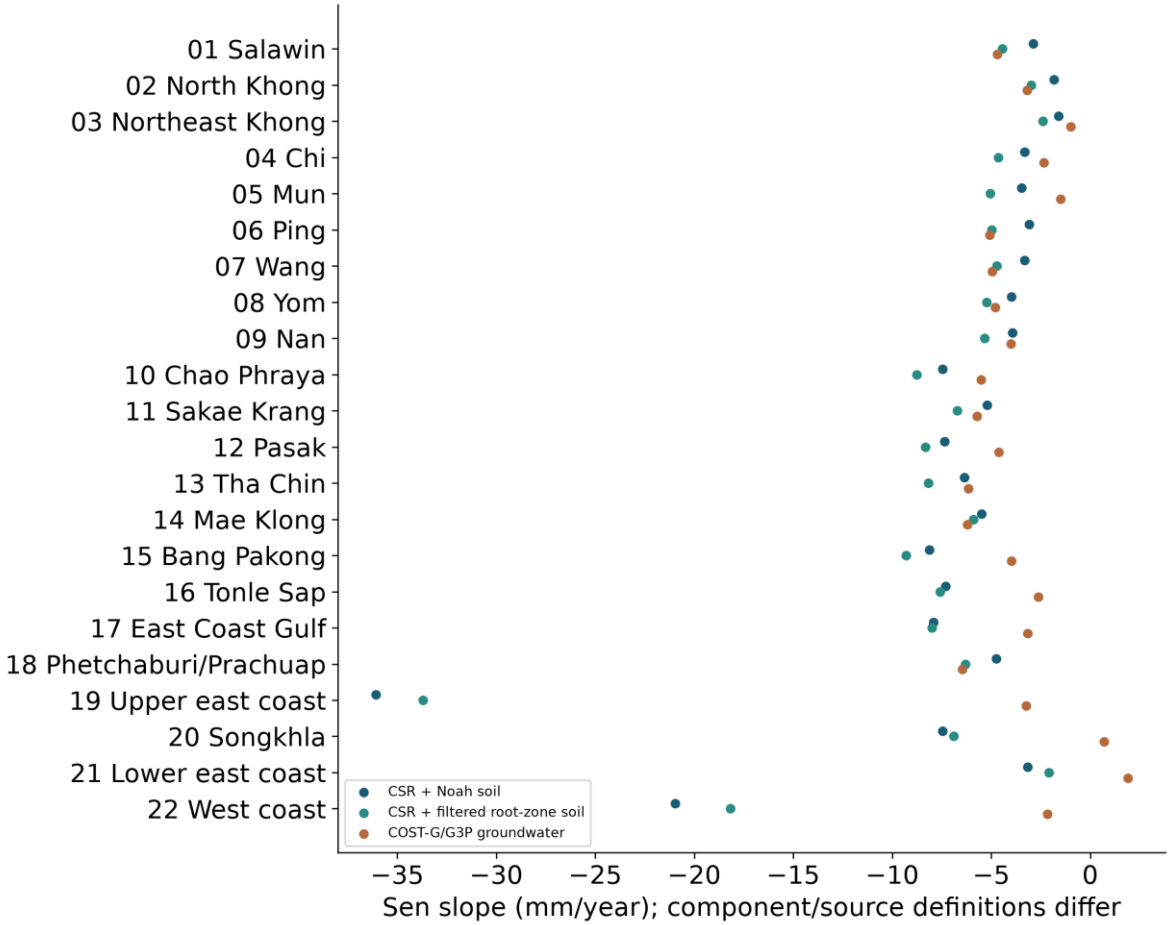


Figure 5.5 Groundwater slopes under alternative estimation structures; differing definitions prevent a single-factor interpretation.

Replacing the primary Noah soil contribution with the filtered root-zone alternative changes the national CSR groundwater slope from −6.84 to −7.86 mm/year. The negative direction remains, but the magnitude changes by about one millimetre per year. The native G3P groundwater chain gives a smaller decline on common source support.

The alternatives do not form an independent probabilistic ensemble. They share satellite information and differ in soil, filtering or gravity-processing choices. Their spread is a structural sensitivity, not a confidence interval or proof that one product is correct. The basin differences show why a detailed local map should retain its product definition and uncertainty context.

Shared native gravity support

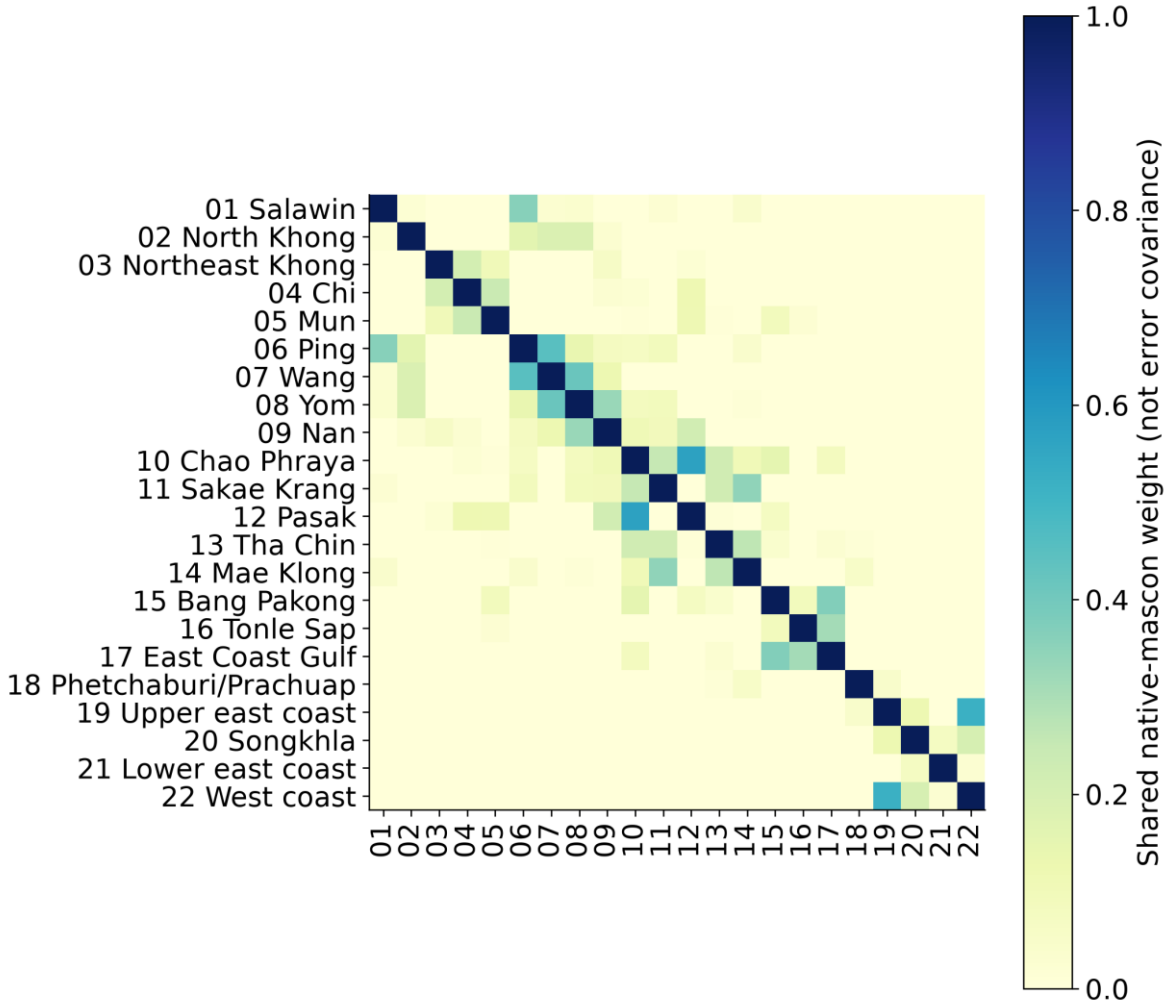


Figure 5.6 Overlap of mapped native-mascon weight fractions; the matrix is not gravity-error covariance.

Some reporting basins share mapped native gravity support. Positive off-diagonal values indicate that part of their spatial representation is drawn from the same native mascon identifiers. The diagonal is one by construction and should not be interpreted as a validation result.

The matrix makes spatial dependence visible without estimating an unavailable full covariance or averaging kernel. Regularization can couple information beyond the mapped overlap. Consequently, the 22 basin tests do not constitute 22 independent satellite experiments. This supports interpreting FDR-adjusted significance and cross-basin rank statistics as screening evidence rather than as complete spatial attribution.

Concentration of basin representation

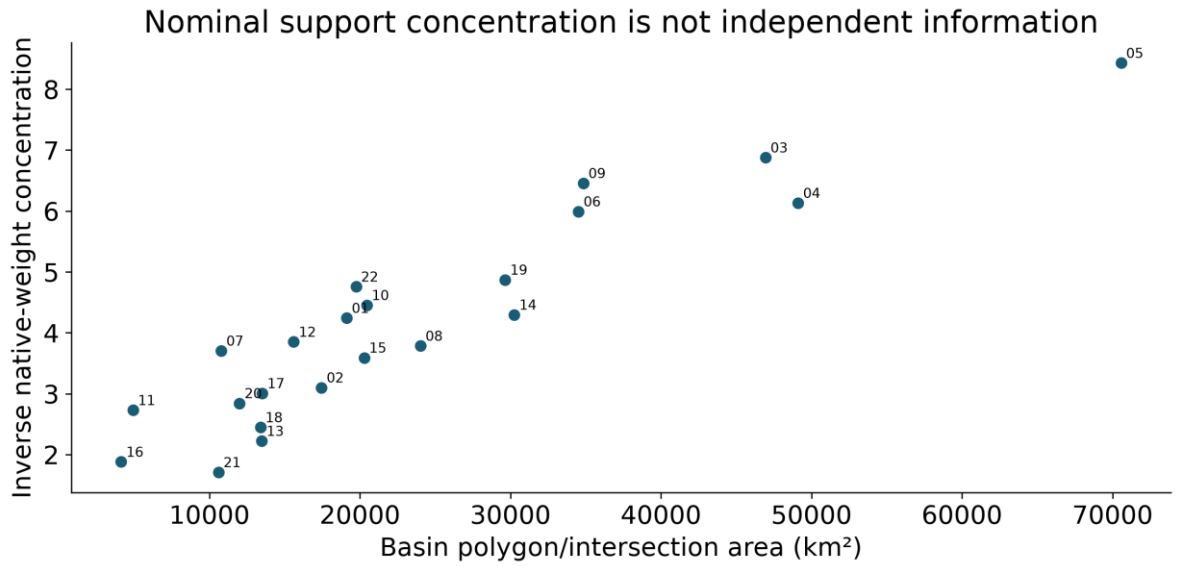


Figure 5.7 Native-weight concentration and basin area; inverse concentration is not an effective statistical sample size.

National context intersects 68 mapped native mascons and has an inverse weight-concentration count near 50.1. Individual basin values are smaller and uneven. A large number of convenience-grid cells therefore does not imply a comparable number of independent gravity observations.

All 22 principal basins remain below the CSR guidance's approximate small-basin caution area. This flag is not a statement that every estimate is unusable. It identifies a scale limitation that should accompany the result. The appropriate use is to combine the storage screen with product agreement, groundwater coverage and independently observed hydrological information, rather than to convert the concentration diagnostic into degrees of freedom.

Climate memory beyond seasonal prediction

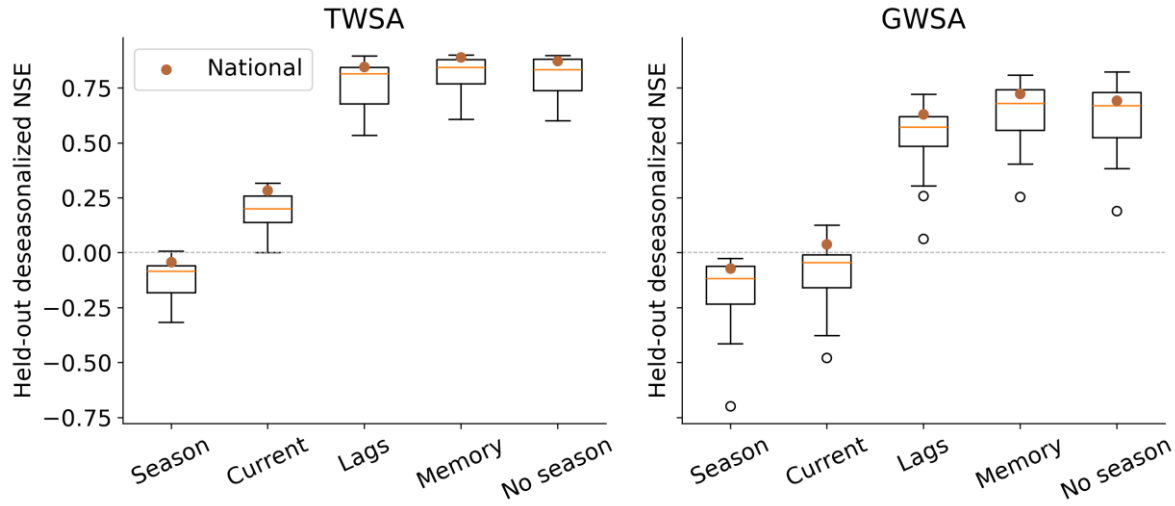


Figure 5.8 Controlled Ridge feature ablations evaluated in 2016–2023; boxes summarize basins and points show national context.

Memory improves reconstruction of anomalies that a seasonal cycle or current weather alone does not capture. National TWSA anomaly NSE rises from 0.283 for current weather to 0.890 with lagged weather and memory. The corresponding GWSA change is from 0.039 to 0.723. The basin-median memory scores are approximately 0.844 and 0.680.

These are parent-model reconstruction results. They demonstrate useful meteorological information and the need to separate seasonality from anomaly skill, but they do not verify actual human depletion. The primary counterfactual remains the published naturalized simulation. The ablation does not replace it merely because a particular statistical feature set scores well.

Sensitivity to independent precipitation forcing

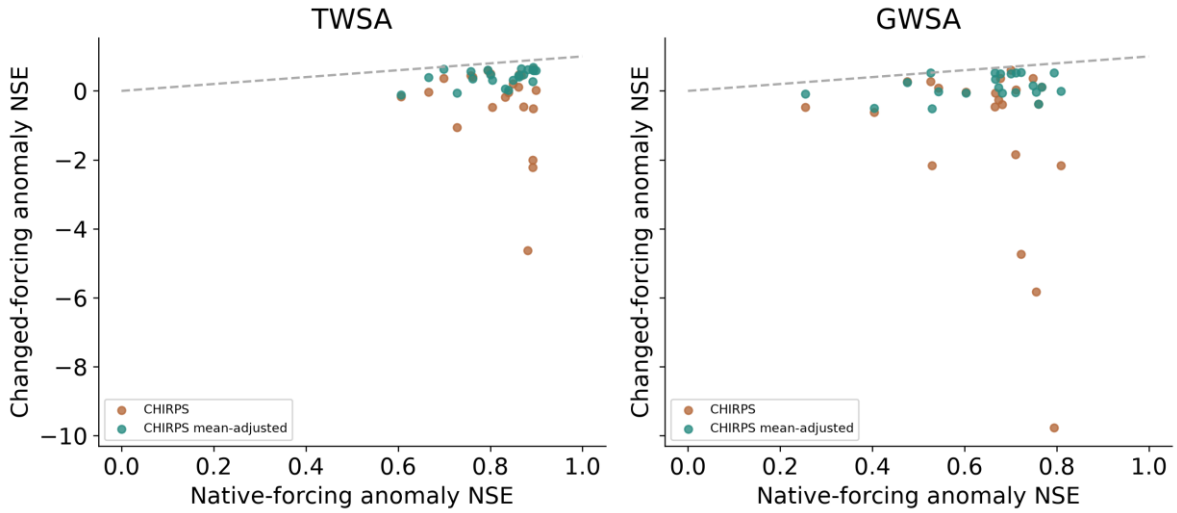


Figure 5.9 Frozen-model forcing replacement against a native-forced naturalized target; this is a sensitivity test, not rainfall-product validation.

Using unadjusted CHIRPS in the frozen memory emulator reduces national anomaly NSE from 0.890 to 0.688 for TWSA and from 0.723 to 0.562 for GWSA. Training-only monthly mean adjustment does not restore the original scores. The test therefore exposes dependence on forcing consistency even when the native reconstruction is strong.

A score reduction does not establish that CHIRPS is less accurate precipitation. The evaluation target was generated with the native forcing. The useful finding is that approximation skill and independent hydrological validity are distinct. Reported performance should identify the input source and target, and model uncertainty should not be represented solely by the spread among algorithms trained on one common parent simulation.

Transfer of observed rainfall and storage associations

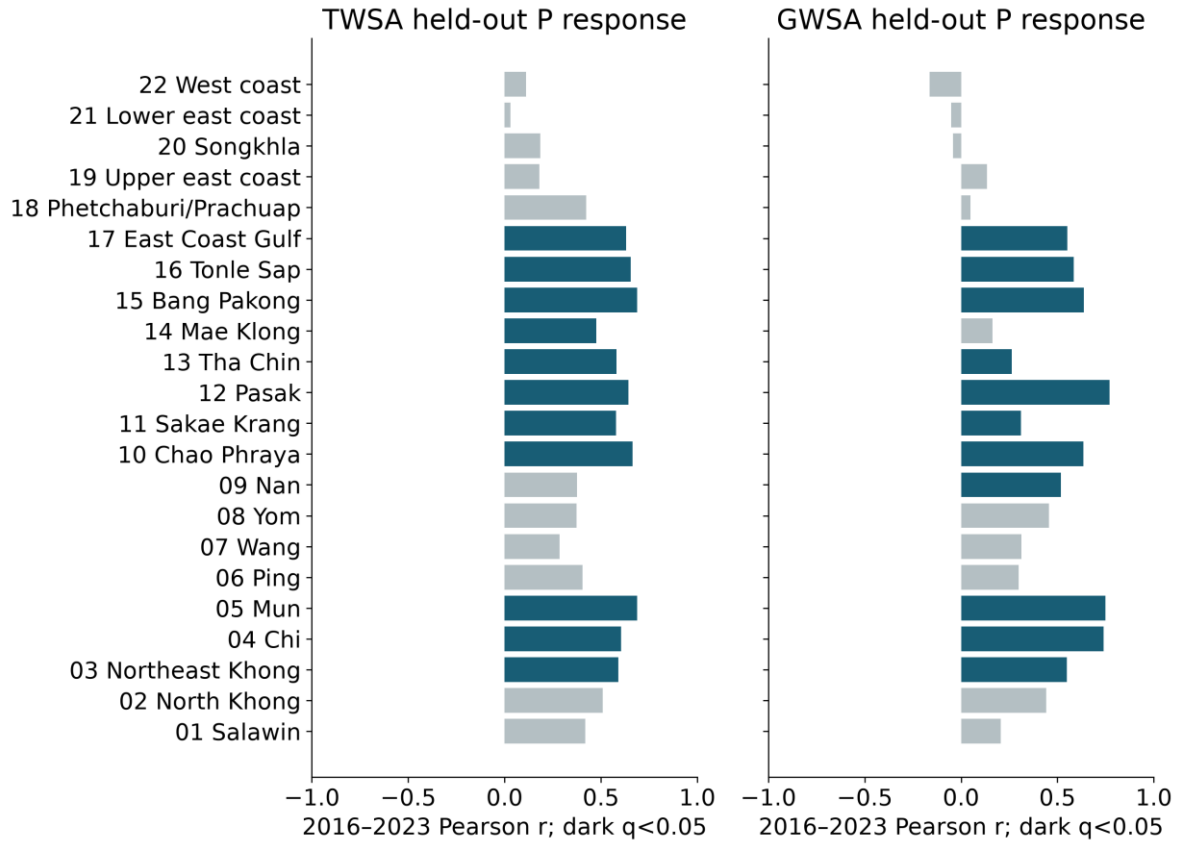


Figure 5.10 Observed 2016–2023 rainfall–storage associations after development-only selection; dark bars pass basin FDR screening.

National observed TWSA is most strongly associated in development with a twelve-month precipitation accumulation delayed by one month; GWSA selects a two-month delay. The held-out correlations are approximately 0.680 and 0.606, based on 79 and 76 paired monthly values. Whole-year precipitation permutations provide a conditional reference for these later-period associations.

The result supports a hydroclimatic role in observed interannual storage variability. It does not make the selected delay an aquifer residence time or identify a purely climatic groundwater response. Irrigation and pumping can also respond to rainfall conditions. The association is therefore interpreted with the paired-model and groundwater evidence, not used to label the remaining trend as automatically anthropogenic.

Paired naturalized and historical human storage

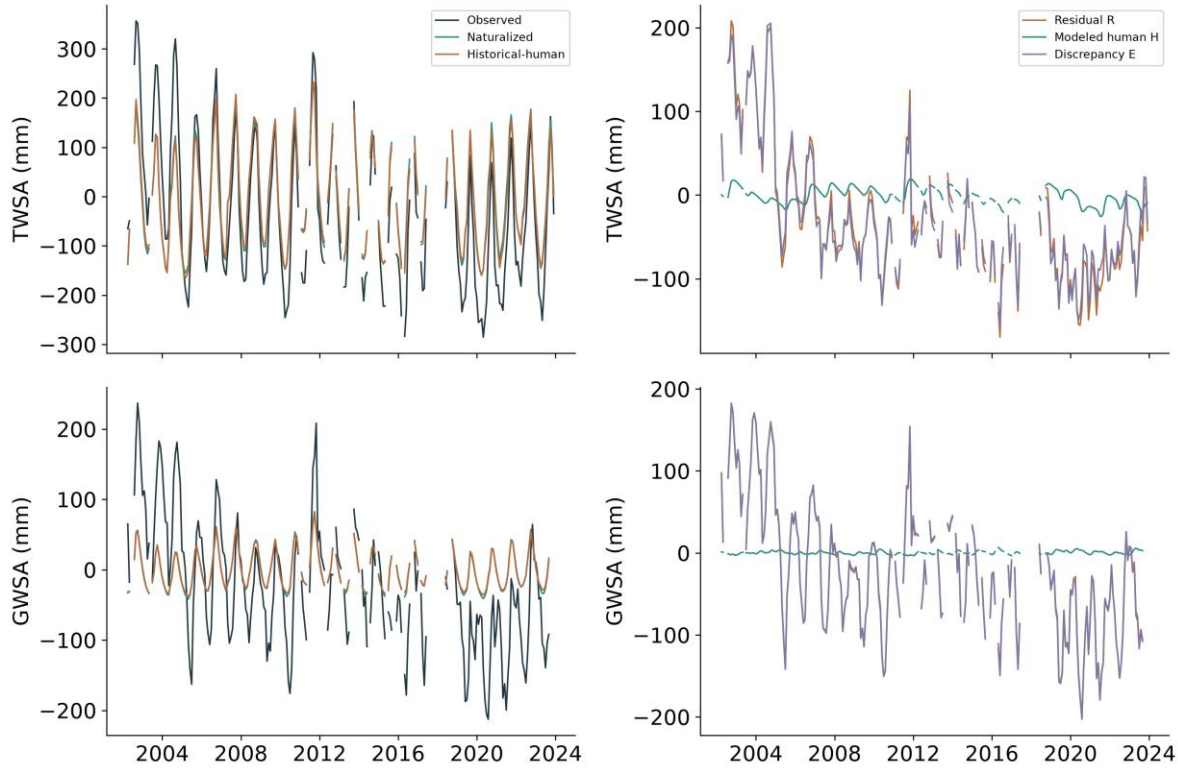


Figure 5.11 Observed and paired modeled storage on common support, with residual R, paired human H and discrepancy E.

Naturalized and historical-human simulations reproduce much seasonal storage behaviour, but substantial observation–model differences remain. The right-hand panels retain both the paired modeled human perturbation and the discrepancy from observed storage. Their sum equals the residual by construction, a relationship verified in the exported data.

A large residual therefore cannot be interpreted as measured human depletion simply because it is calculated after subtracting a naturalized baseline. The historical-human simulation does not reproduce that residual fully. This leaves an unresolved combination of unrepresented management, process and forcing errors, and observation or compartment effects. The experiment provides structured attribution evidence without uniquely assigning those alternatives.

Additive trend decomposition

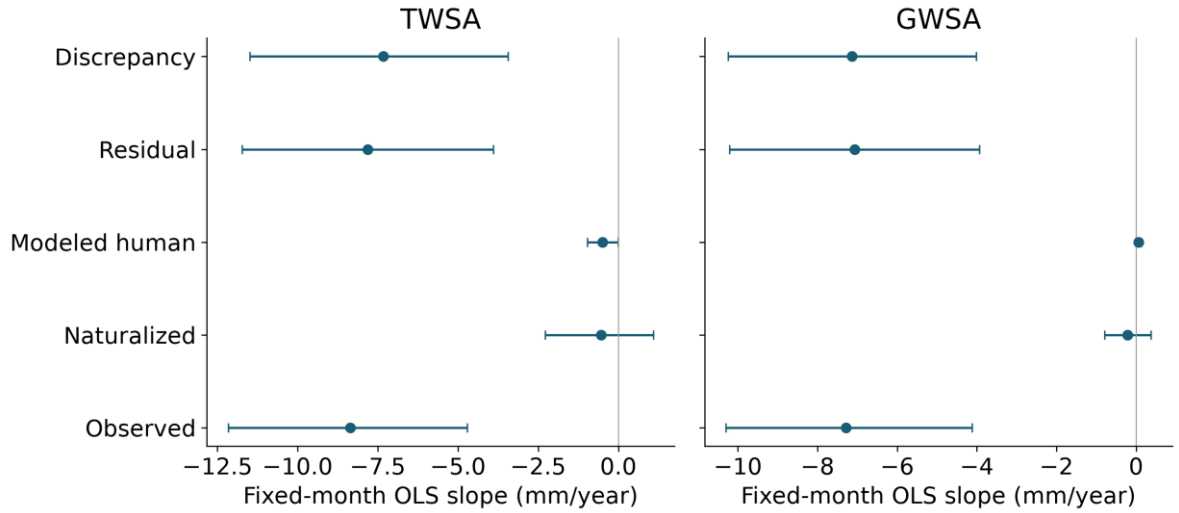


Figure 5.12 Monthly-fixed-effect OLS slopes with shared two-year-block wild-bootstrap intervals; these point slopes are additive.

The common-date national TWSA OLS slopes are −8.36 mm/year observed, −0.54 naturalized, −0.49 paired modeled human, −7.81 residual and −7.32 discrepancy. The corresponding groundwater values are −7.28, −0.21, +0.06, −7.07 and −7.13 mm/year. These coefficients are distinct from the robust Sen summaries and preserve the exact component identities.

For groundwater, the modeled-human two-year-block interval spans approximately −0.01 to +0.13 mm/year, while the residual and discrepancy intervals remain negative. This does not prove the discrepancy is non-human error. It shows that the represented paired human perturbation does not account for the observed residual trend within this experiment.

Dependence on uncertainty block length

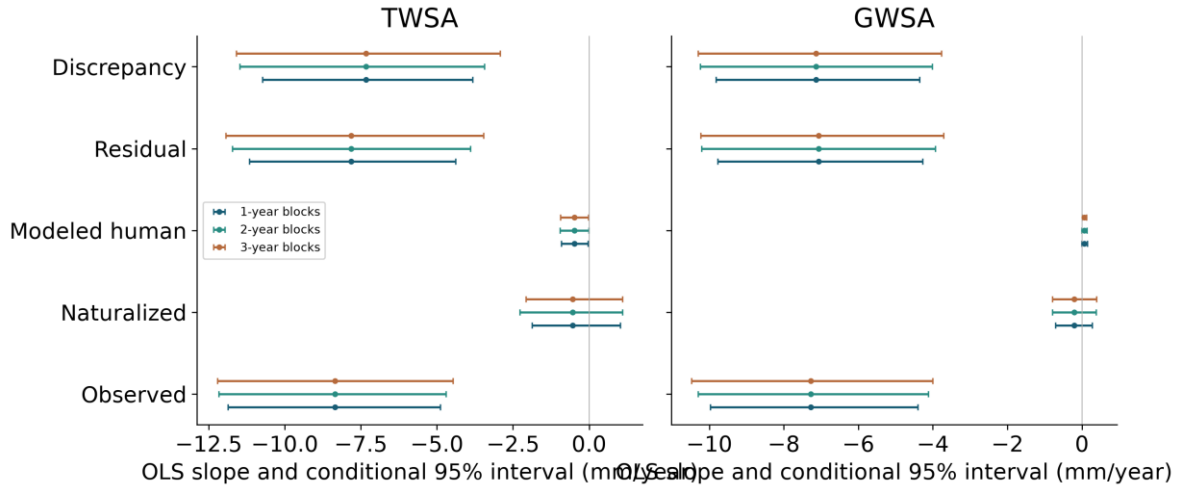


Figure 5.13 Conditional slope intervals under one-, two- and three-year calendar blocks; missing dates are retained.

Longer blocks permit more persistence in the regression residuals and can alter uncertainty around small component trends. The interval comparison is particularly relevant to the modeled-human term, whose magnitude is much smaller than the observed residual. A borderline interval under one block choice should not be treated as decisive attribution evidence.

Shared multipliers preserve the decomposition in every replicate. The intervals remain conditional on the chosen model, processed inputs and linear trend specification. They do not propagate complete GRACE inversion uncertainty or the full range of plausible groundwater-model structures. Product and compartment comparisons therefore remain separate parts of the evidence rather than being absorbed into a single narrow interval.

Discrepancy magnitude and attribution credibility

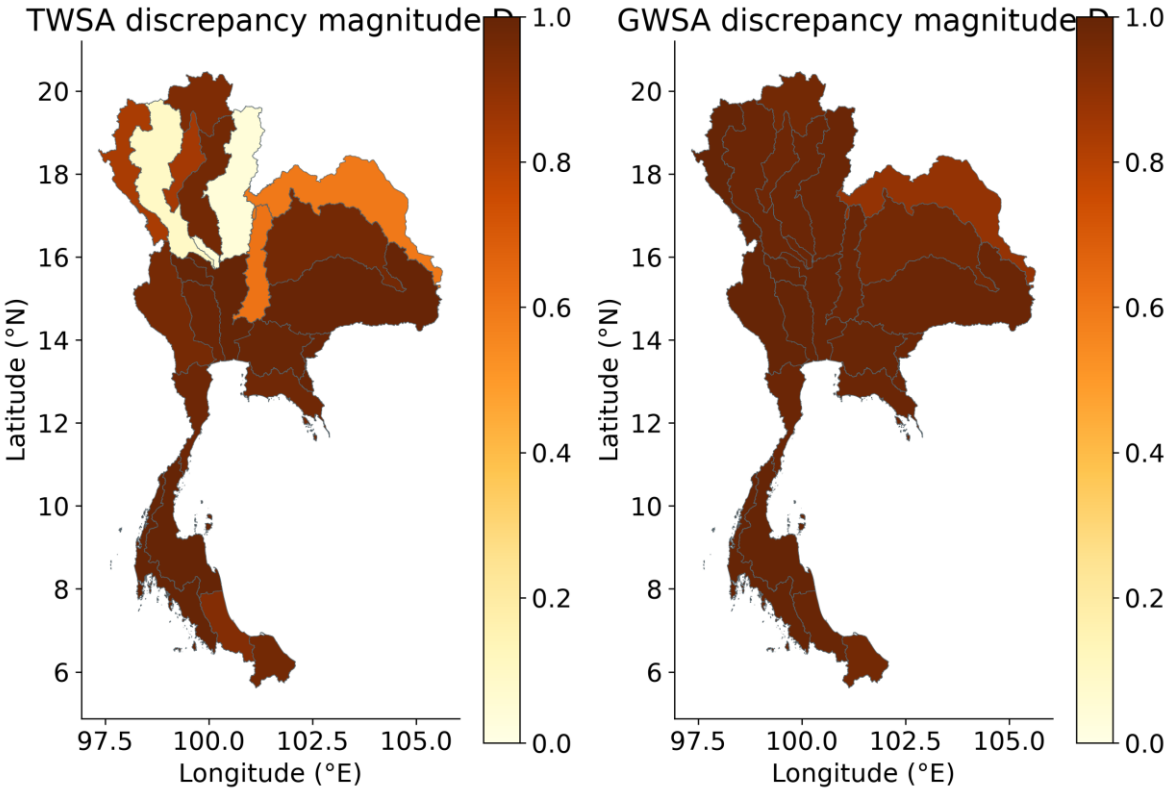


Figure 5.14 Descriptive discrepancy-magnitude index D; it is not a human fraction or an independent error-budget allocation.

The national two-year-block diagnostic is 0.938 for TWSA and 0.991 for GWSA. Values near one indicate that the discrepancy trend is large relative to the paired modeled-human trend in this magnitude comparison. The map exposes where the interpretation of an observed residual depends strongly on the unexplained part of the historical-human simulation.

Neither a high index nor its conditional interval identifies the cause of the discrepancy. Omitted human intervention and hydrological-model error are both possible contributors. The diagnostic should be used to qualify an attribution claim and guide further investigation, not to estimate a proportion of depletion caused by humans. This distinction is retained in the manuscript and decision-support material.

Observation network coverage through time

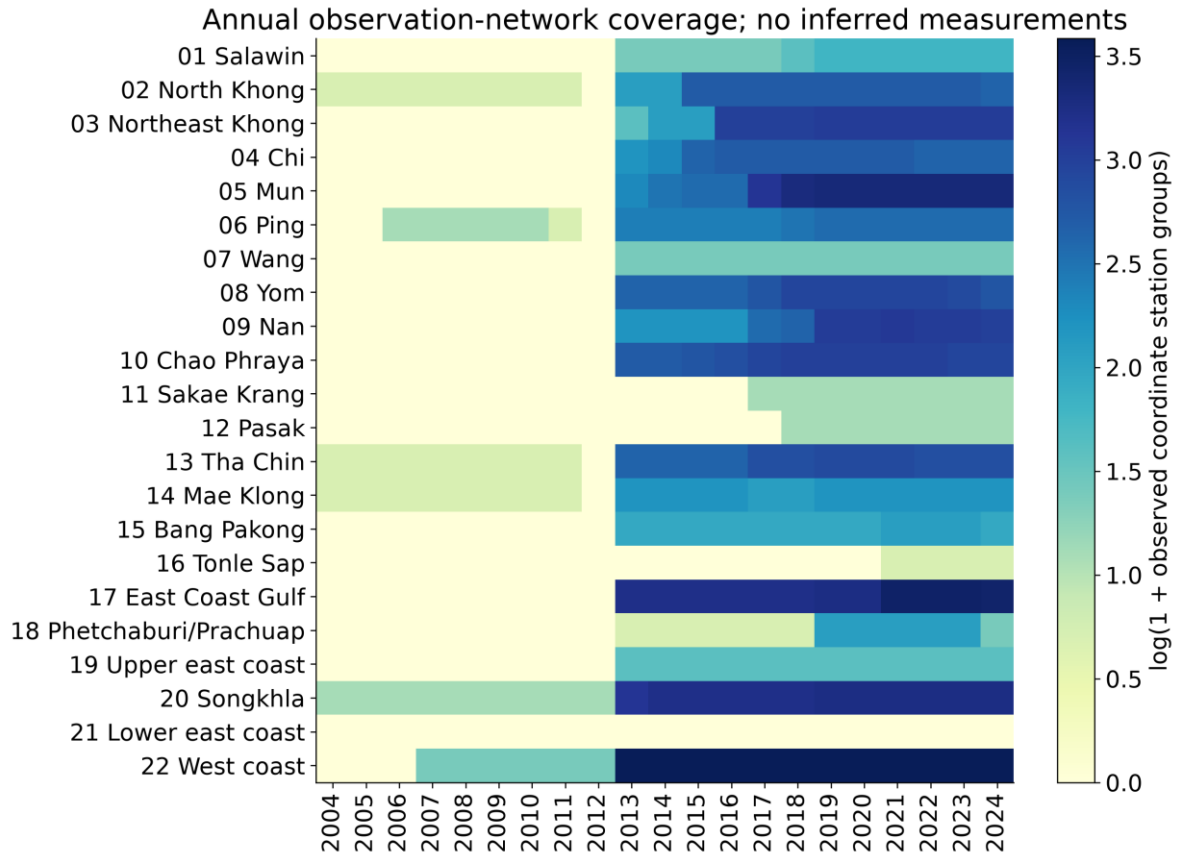


Figure 5.15 Annual coordinate-station availability by basin; values display log of one plus actual station count.

The well dataset is spatially and temporally uneven despite its large total number of records. Annual coverage changes materially through the observation period, and some basins lack a useful annual validation population. Co-location grouping and the coverage matrix expose the difference between the number of well identifiers and the number of contributing station groups.

This changing network can influence a national or basin head index even when every retained observation is real. The study therefore retains annual counts, source flags and panel definitions. A smooth index is not evidence that the same aquifers were sampled throughout the record. Missing or weakly supported comparisons are reported explicitly rather than replaced with fabricated correlations.

Common period and balanced panel comparisons

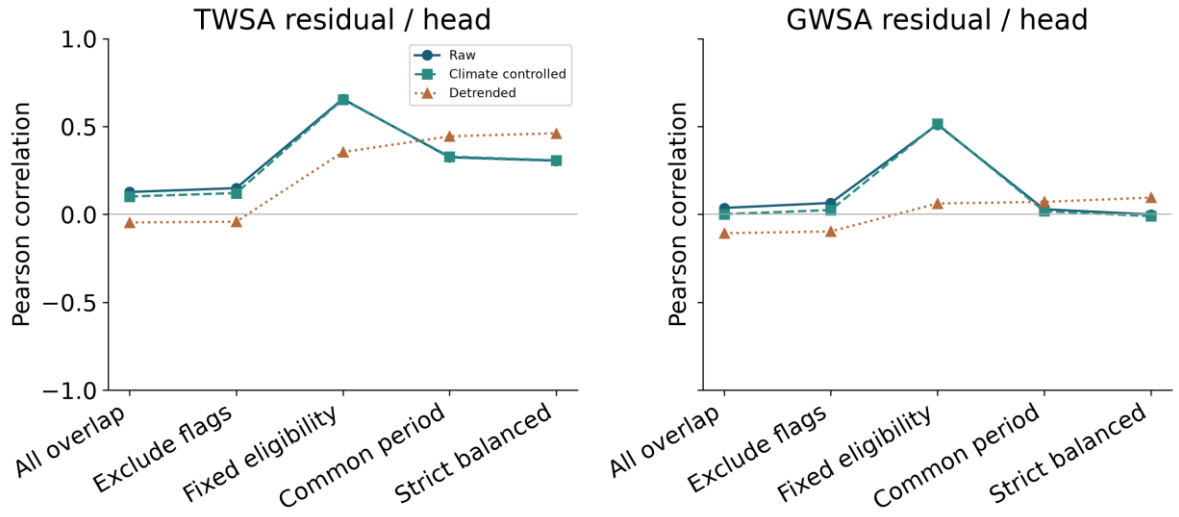


Figure 5.16 Residual–head relationships across quality and panel conditions; different time supports must be distinguished.

The longest-overlap national residual–head correlations are 0.127 for TWSA and 0.037 for GWSA. The fixed-eligibility, wider-history panel gives larger raw values, approximately 0.657 and 0.511. That improvement cannot be interpreted as validation without separating a panel effect from a time-support effect.

On the common 2013–2022 window, only eight storage years pass the availability rule. The all-well GWSA residual correlation is 0.029; the strict balanced panel gives approximately −0.0004. The latter uses 189 coordinate station groups. Groundwater agreement therefore remains weak under the fairer time-matched comparison, despite an apparently attractive correlation under the broader fixed-eligibility condition.

Table 5.2 National GWS residual and head comparisons

Panel	Years	Raw r
All overlap	17	0.037
Fixed eligibility	17	0.511
All wells common period	8	0.029
Strict balanced	8	-0.000

Well head trajectories and direction uncertainty

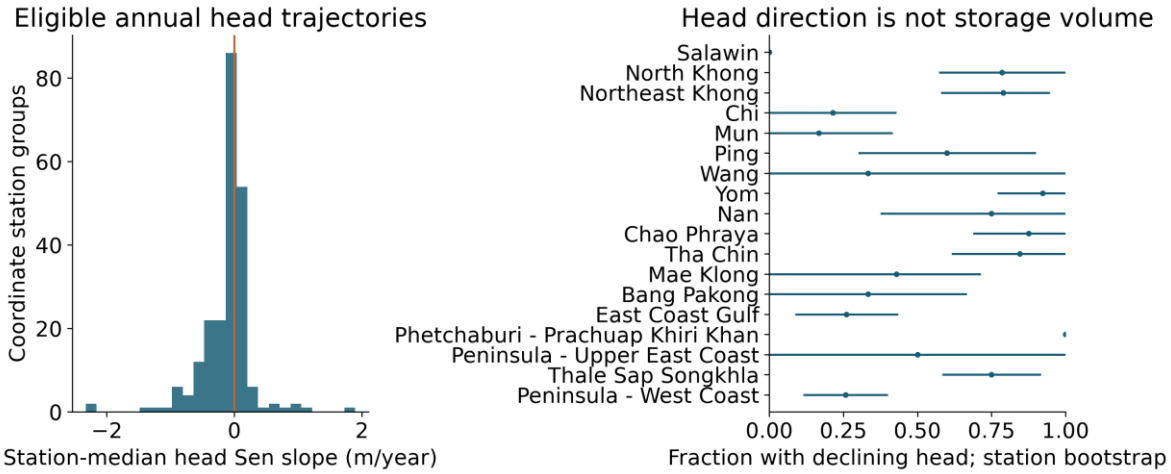


Figure 5.17 Eligible annual head trends and station-resampled declining fractions; head units are not storage units.

The well-level trend screen retains 382 annual wells meeting the observation and span criteria. After co-location grouping, 225 station trajectories contribute to the national direction summary. The median station head slope is approximately −0.016 m/year, and 54.2 percent of station-median slopes are negative.

The station-resampling interval for the declining fraction is about 47.1 to 60.0 percent. It does not establish a clearly dominant national direction within the monitored population. More importantly, the population is not a random sample of all Thai aquifers. Head trajectories provide useful local evidence, but their magnitude cannot be equated with millimetres of regional GWSA without verified storage coefficients and aquifer support.

Basin groundwater agreement and station uncertainty

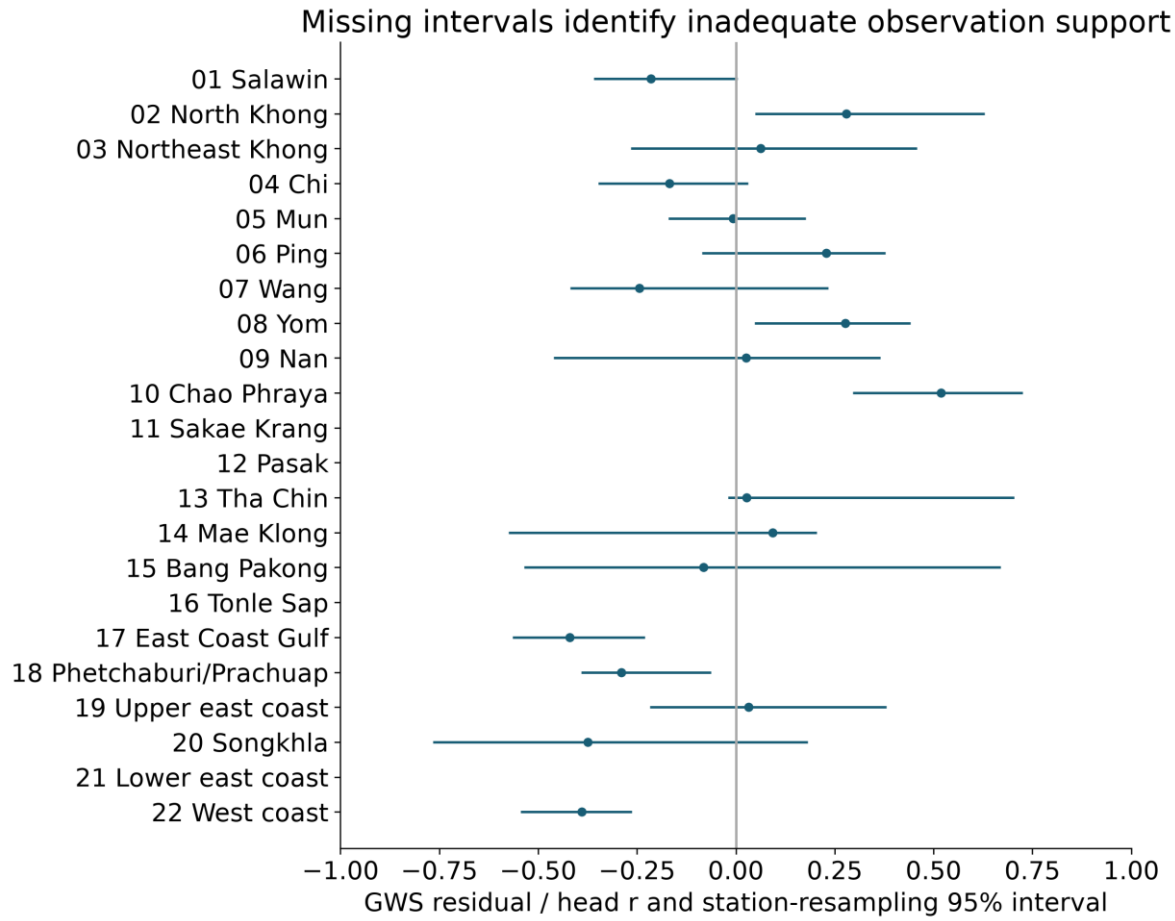


Figure 5.18 Primary GWS residual–head associations and coordinate-station bootstrap intervals; absent intervals indicate inadequate support.

Basin-level correlations vary, but their interpretation depends on the number and distribution of station groups and the common observed years. The station bootstrap holds those years fixed while resampling the monitored spatial population. Its intervals therefore describe network sensitivity, not all temporal or product uncertainty.

A basin with a positive point correlation is not automatically validated, particularly where few independent station groups or few annual pairs are available. A basin with no adequate comparison is an observation gap, not evidence of either absence or presence of depletion. The decision-support interpretation retains this distinction and directs attention to the monitoring information needed for a stronger assessment.

Trend windows and shorter period recoveries

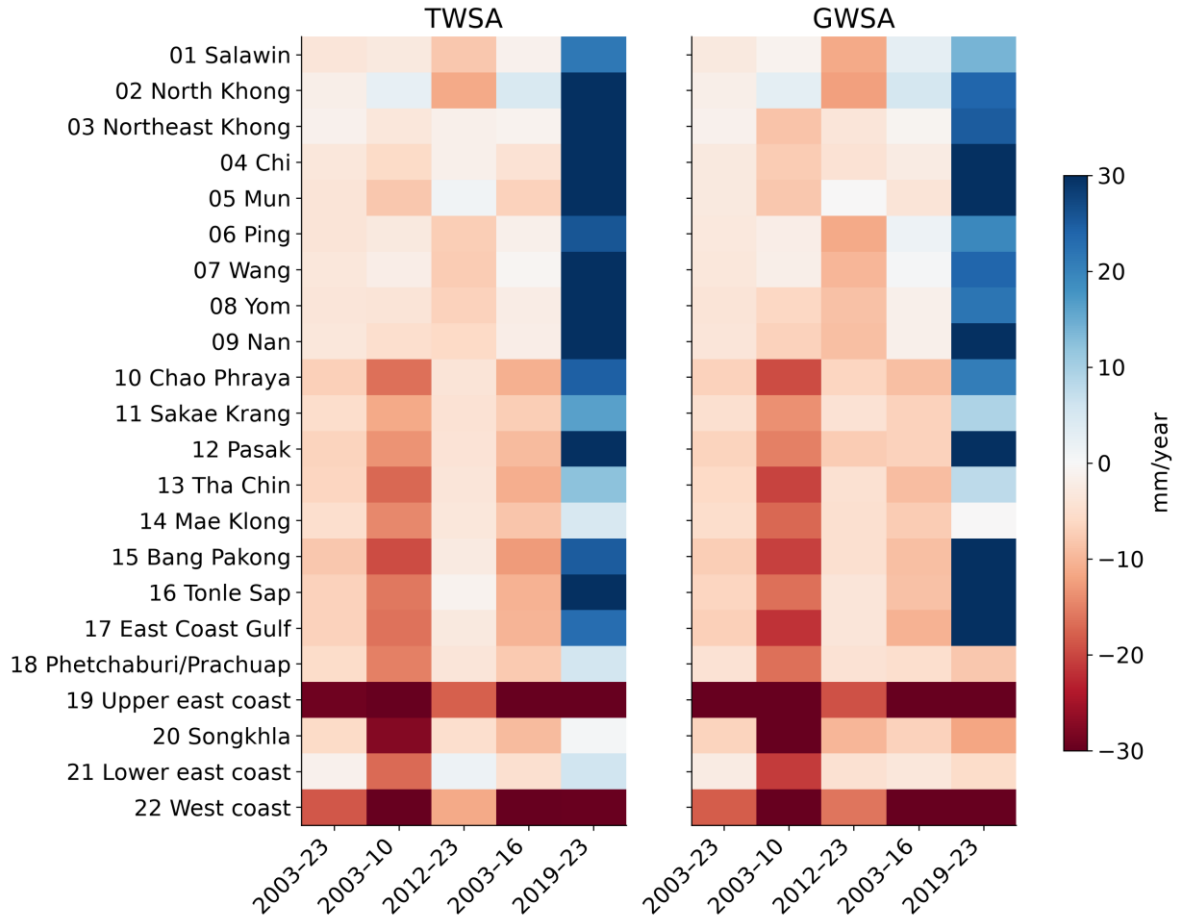


Figure 5.19 Fixed-window seasonal Sen slopes; colours saturate at ±30 mm/year and full numerical values remain in the CSV.

The record is not a monotonic trajectory. Several later-period basin slopes are positive even where the long common-record trend is negative. In contrast, some southern records retain strong negative slopes across windows. The fixed comparisons reveal the influence of period choice without selecting a window solely because it produces a desired direction.

The 2019–2023 window is short and can be dominated by a few wet or dry years. Its slope should not be interpreted as a forecast or as evidence of permanent recovery. The same caution applies to steep early-window declines. Longer records, source comparisons and independent compartment observations are needed to distinguish transient variation from a durable change in storage regime.

CHAPTER 6
DISCUSSION

Interpreting robust decline and unstable local rankings

The national and basin results answer different management questions. A consistent negative national trend across processing chains supports concern about broad storage change. The much weaker agreement in basin ranking shows that a national conclusion cannot be translated directly into a reliable ordering of local severity. This is especially important when administrative boundaries are smaller than the effective support of the gravity inversion.

The close relationship between the unidentified export and the documented CSR record is reassuring as a numerical audit but does not prove the export's origin. It should not be counted as an additional independent gravity product. The more informative sensitivity is the documented contrasting COST-G/G3P chain. Its differences show why processing and compartment structure belong in the interpretation of magnitudes and local patterns.

Groundwater subtraction introduces an additional layer of dependence. Changing the soil representation alters the national groundwater slope, while using the complete G3P chain changes both magnitude and ranking. The resulting spread is not a formal probabilistic uncertainty envelope because the products are related and do not span all plausible structures. It is evidence that conclusions requiring an exact depletion rate are less secure than conclusions about broad direction.

The short-window analysis also limits a simple narrative of continuously worsening conditions. Later wet intervals can produce positive local slopes even when the full record declines. Such behaviour is physically plausible in a climate-sensitive storage system and should not be used to erase the long-record signal. Conversely, a short steep decline should not be extrapolated as a constant future loss rate.

For basin prioritization, the appropriate response is to combine the trend with observation support, product agreement and independent compartment evidence. The present data justify identifying candidate areas for investigation. They do not justify precise local pumping quantities or a product-independent ranking of the most affected aquifers. This distinction is a substantive finding of the national assessment, not a qualification added after the results.

Why strong climate reconstruction does not validate human depletion

The climate experiments demonstrate that antecedent meteorology contains substantial information about naturalized storage. Lagged and memory features outperform a seasonal or current-weather description, including for deseasonalized anomalies. This is a useful result for developing compact climate-storage reconstructions. It also shows why models should be evaluated beyond a recurring annual cycle rather than judged by raw correlation alone.

The forcing experiment places that success in context. Replacing native precipitation with CHIRPS changes anomaly skill even though the statistical model is frozen. This does not rank the rainfall products by observational accuracy. It demonstrates dependence on the input–target relationship of the parent simulation. An emulator that reproduces a model accurately can inherit that model's structural limits and forcing dependence.

The paired attribution experiment exposes a separate problem. Most of the observed residual trend remains in the observation–historical-model discrepancy. The modeled direct-human perturbation is much smaller and, for groundwater, its conditional interval includes zero under the principal two-year block choice. This result is incompatible with interpreting every residual unit as a validated human contribution within the present model experiment.

The discrepancy can contain actual management omitted from the model, as well as forcing error, groundwater process error and mismatched storage definitions. The analysis cannot distinguish these explanations uniquely. It would be equally unjustified to call the entire discrepancy non-human error. The defensible interpretation is unresolved attribution, with the residual retained as an observation-informed candidate signal rather than a measured pumping trajectory.

Even climate-controlled head correlation has a mathematical limitation. Because R equals O minus C, controlling the same C in O and R produces the same residualized storage vector. A similar partial correlation for observed storage and its residual is therefore expected from the identity and is not independent proof that climate has been perfectly removed. The groundwater evidence must be assessed through its actual support and robustness, not through the name assigned to the conditioning variable.

Monitoring priorities and the proposed novelty

The station-panel analysis provides a concrete warning about validation design. A fixed-eligibility index can show a stronger raw association while still changing its represented population through time. The common-period and strictly balanced comparisons are more informative about that issue. Their groundwater correlations remain close to zero in this study, indicating that the apparent improvement does not establish a validated human-depletion component.

An exploratory independent surface-water check uses MSEA-Res Landsat and hypsometric reservoir volumes (Mahto et al., 2025). Of 27 Thai reservoirs located within the basin polygons, 26 were commissioned before 2004 and form a fixed comparison panel. National TWS residual correlation is 0.521 across 73 scene-supported months and 0.439 after removing calendar-month effects and linear trend; a 499-replicate year-cluster interval is 0.188–0.665. Provider-interpolated Level-2 storage gives 0.381 adjusted correlation across 228 months. This supports surface-water consistency, not causal human attribution: reservoir volumes respond to both climate and operations, and no naturalized reservoir counterfactual is observed.

The proposed novelty has three connected parts. First, the study separates stable national direction from unstable basin ranking across documented processing chains and shared native support. Second, it demonstrates the value and limits of climate-memory emulation through controlled ablation and forcing replacement. Third, it shows how common-period, balanced-panel and station-bootstrap comparisons alter the credibility of groundwater validation.

These contributions are methodological integration and empirical evidence. The study does not claim that the accounting identity, Ridge regression or a bootstrap is a new algorithm. Its added value is an auditable national experiment in which favourable and unfavourable evidence are reported together. The public interface and executable package make those inferential links reviewable rather than asking readers to trust a single depletion map.

For decision support, the results should guide investigation and data collection. Basins with strong decline but weak product agreement warrant source and compartment checks. Basins with sparse wells warrant observation-network improvement. A high emulator score can support reconstruction of a parent model, while management interventions require stronger causal and volumetric evidence. This evidence-based use is more defensible than converting an uncertain residual directly into a withdrawal restriction.

CHAPTER 7
CONCLUSIONS AND RECOMMENDATIONS

Conclusions and research priorities

The study completes a national basin assessment of terrestrial and groundwater storage, a controlled climate-storage reconstruction and attribution experiment, and a reproducible decision-support release. Its primary gravity baseline is now identified as CSR RL06.3, with source checksums, land masking and an audited monthly calendar. The full-compartment groundwater calculation and paired residual identities are verified numerically.

Observed national storage declines in the analyzed record. The direction is also negative in the documented contrasting product chain, but the magnitude and basin ordering are product-dependent. All 22 principal basins retain a small-support caution. Spatial maps should therefore be read as screening evidence, and the national conclusion should not be converted into a precise ranking of local abstraction stress.

Climate-memory features improve reconstruction of naturalized model storage beyond seasonal and current-weather baselines. Their success persists in held-out anomalies, while independent precipitation replacement exposes forcing sensitivity. These results support a useful meteorological reconstruction framework. They do not establish that the parent model provides an exact natural counterfactual or that all unexplained observed change is human-caused.

The paired human experiment and the groundwater analyses limit that attribution claim. The observation–historical-model discrepancy dominates the residual trend diagnostic. Longest-overlap and balanced-panel groundwater checks provide weak support for the derived residual, and the station bootstrap exposes substantial observation-network sensitivity. A consistent negative head trend at some stations is evidence about those trajectories, not a calibrated estimate of regional groundwater mass loss.

Further progress requires verified aquifer information, representative monthly heads, storage coefficients, pumping histories, reservoir observations and additional naturalized model structures. The thesis and manuscript provide a completed research draft and an explicit set of testable findings for advisor and peer review. Their contribution is the evidence-qualified assessment, including unresolved attribution, rather than a guarantee of journal placement or a predetermined explanation of the decline.

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APPENDIX A
ALL BASIN RESULTS

Table A.1 All-basin primary slopes and mapped support

Basin	TWSA mm/yr	GWSA mm/yr	Native conc.
01 Salawin	-3.59	-2.88	4.2
02 North Khong	-1.90	-1.83	3.1
03 Northeast Khong	-1.32	-1.59	6.9
04 Chi	-3.65	-3.32	6.1
05 Mun	-4.42	-3.47	8.4
06 Ping	-3.71	-3.10	6.0
07 Wang	-3.21	-3.32	3.7
08 Yom	-3.79	-3.97	3.8
09 Nan	-3.74	-3.94	6.5
10 Chao Phraya	-7.36	-7.47	4.5
11 Sakae Krang	-5.50	-5.20	2.7
12 Pasak	-7.06	-7.35	3.9
13 Tha Chin	-6.89	-6.36	2.2
14 Mae Klong	-5.25	-5.48	4.3
15 Bang Pakong	-8.64	-8.12	3.6
16 Tonle Sap	-7.68	-7.30	1.9
17 East Coast Gulf	-7.51	-7.92	3.0
18 Phetchaburi - Prachuap Khiri Khan	-5.85	-4.75	2.4
19 Peninsula - Upper East Coast	-34.37	-36.08	4.9
20 Thale Sap Songkhla	-6.25	-7.45	2.8
21 Peninsula - Lower East Coast	-1.62	-3.16	1.7
22 Peninsula - West Coast	-21.48	-20.95	4.8

Native concentration is inverse weight concentration, not an independent sample count. All principal basins retain the provider small-support caution. Complete tests, intervals, coverage and groundwater evidence are in the versioned CSV tables.

APPENDIX B
REPRODUCIBILITY AND INTERPRETATION CONTRACT

The extended numerical study is implemented in numbered acquisition, storage, robustness, climate, groundwater and synthesis scripts. Compact regional inputs and their provenance are retained. The documented CSR record is a new primary input, while the earlier supplied export remains a sensitivity. No original research output is silently overwritten; the extended study occupies its own versioned directory.

A reproducible run verifies input checksums, rebuilds the scientific tables and figures, and checks the accounting identities, model split rules, observation counts and table schemas. Frozen reference outputs are used for comparison only. They are not substituted for recomputed scientific results. The repository documentation distinguishes the original executable release from the extended analyses and records the environment used.

Interpretation follows the same contract as execution: groundwater heads remain heads, model-emulation skill remains model-emulation skill, and a residual remains the sum of the represented human perturbation and unresolved discrepancy. Conditional bootstrap intervals do not become full product uncertainties, and native support concentration does not become a count of independent observations. Keeping those meanings intact is necessary for a reader to reproduce the scientific argument as well as the numerical tables.

The accompanying manuscript focuses on product dependence, climate memory and network uncertainty; the thesis provides the fuller data, methods and basin context. The Beamer source and presentation summarize the same checked results. All author information uses Farzeen Khan, with academic supervision acknowledged to Dr. Natthachet Tangdamrongsub. Original dataset attributions and source-specific reuse terms remain attached to the released material.

Code and data access: https://github.com/pratikkhanal16/thailand-water-storage-dss